

ON THE CONVERGENCE OF THE ORTHOGONAL SPECTRAL SEQUENCE

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Doctoral Thesis — Universidad Nacional Autónoma de México
October 2022

Abstract

One of the central theorems in Algebraic Geometry is the Néron–Severi theorem, which asserts that for a smooth projective variety X over an algebraically closed field, the group $\mathrm{CH}^1(X)/\mathrm{CH}_{\mathrm{num}}^1(X)$ is finitely generated. Motivated by this, Bloch, Beilinson, and Murre independently introduced a conjectural filtration on the higher Chow groups $\mathrm{CH}^r(X)$, $r > 1$. To study finite filtrations on these groups, Peláez introduced in [28] a tower of triangulated functors $bc_{\leq \bullet}$ in the Voevodsky category of mixed motives DM and associated a spectral sequence

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(B, A).$$

The main object of this thesis is to study the convergence of this *orthogonal spectral sequence*. The main result (Theorem 2.13) shows that this spectral sequence converges strongly when $B = M(X)(r)[s]$, with $r, s \in \mathbb{Z}$ and X any smooth scheme of finite type over k . We also show that the analogous result fails in the $\mathbb{A}^1$ -stable homotopy category $\mathcal{SH}$ of Morel–Voevodsky (Proposition 2.16), and we identify conditions on $A \in \mathcal{SH}$ under which strong convergence holds (Proposition 2.19). As a direct consequence, we obtain a spectral sequence that converges strongly at the E_1 -term of the Voevodsky slice spectral sequence (Corollary 2.20).

Acknowledgements

The author would like to thank Dustin Clausen for illuminating conversations, and Chuck Weibel for his interest in this work.

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Introduction

One of the central theorems in Algebraic Geometry is the *Néron–Severi theorem* [29, 25, 26]: for a smooth projective variety X over an algebraically closed field, the group $\mathrm{CH}^1(X)/\mathrm{CH}_{\mathrm{num}}^1(X)$ is finitely generated, where $\mathrm{CH}^1(X)$ is the group of algebraic cycles of codimension 1 modulo rational equivalence and $\mathrm{CH}_{\mathrm{num}}^1(X)$ is the subgroup of numerically trivial cycles. Since $\mathrm{CH}_{\mathrm{num}}^1(X)$ is the group of rational points of the Picard variety of X (an abelian variety), this result decomposes $\mathrm{CH}^1(X)$ into a continuous component — parametrised by an abelian variety — and a discrete component $\mathrm{CH}^1(X)/\mathrm{CH}_{\mathrm{num}}^1(X)$ which is a finitely generated abelian group.

Motivated by this, the expectation was that for the higher Chow groups $\mathrm{CH}^r(X)$, $r > 1$, there would similarly exist a subgroup parametrised by some naturally associated algebraic variety such that the quotient was finitely generated. However, this was dashed by Mumford [22], who proved that for a surface S with $\dim H^0(S, \Omega_S^2) > 0$, the group of zero cycles $\mathrm{CH}^2(S)$ is “enormous” and in particular admits no such parametrisation.

This led experts to conclude that the Chow groups $\mathrm{CH}^r(X)$ for $r > 1$ are in general chaotic. Nevertheless, in the late 1970s Bloch, Beilinson, and Murre independently introduced [4, 3, 15] a conjectural programme in which the algebraic cycle groups $\mathrm{CH}^r(X)$, $r > 1$, have accessible properties. The central piece of this programme is the existence of a filtration on $\mathrm{CH}^r(X)$ whose associated graded pieces have accessible properties.

In order to study finite filtrations on the Chow groups of a smooth projective variety Y over a field k satisfying some properties of the (still conjectural) Bloch–Beilinson–Murre filtration, Peláez introduced in [27] a tower of triangulated functors $bc_{\leq \bullet}$:

$$\cdots \rightarrow bc_{\leq n-1} \rightarrow bc_{\leq n} \rightarrow bc_{\leq n+1} \rightarrow \cdots$$

in the Voevodsky triangulated category of mixed motives DM . The filtration on the Chow groups with coefficients in a commutative ring R is defined by evaluating the orthogonal tower $bc_{\leq \bullet}$ on the motive of a point $\mathbb{Z}_R$ and then considering morphisms $M(Y)(-q)[-2q] \rightarrow bc_{\leq \bullet}(\mathbb{Z}_R)$, where $M(Y)$ is the motive of Y and $(-q)$ (resp. $[-2q]$) denotes the Tate twist in DM (resp. the suspension in DM). The previous process provides a filtration on the Chow groups since

$$\mathrm{CH}^q(Y)_R \cong \mathrm{Hom}_{DM}(M(Y)(-q)[-2q], \mathbb{Z}_R).$$

Given $A, B \in DM$, it is possible to evaluate the tower on A and then map B into $bc_{\leq \bullet}(A)$ to obtain a spectral sequence:

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(B, A), \quad (1)$$

where $bc_{p/p-1}A$ is defined in terms of a canonical distinguished triangle in DM

([28], 3.2.8):

$$bc_{\leq p-1}A \rightarrow bc_{\leq p}A \rightarrow bc_{p/p-1}A.$$

The main goal of this thesis is to study convergence properties of this spectral sequence. The main result (Theorem 2.13) shows that the spectral sequence (1) converges strongly for $B = M(X)(r)[s]$, with $r, s \in \mathbb{Z}$ and X any smooth scheme of finite type over k .

It is worth pausing on why convergence is not automatic. The spectral sequence (1) is built, exactly as in the classical Postnikov-tower or Adams spectral sequence construction in stable homotopy theory (cf. [13]), from an *infinite* tower of triangles $bc_{\leq p-1}A \rightarrow bc_{\leq p}A \rightarrow bc_{p/p-1}A$ indexed by $p \in \mathbb{Z}$, rather than from a bounded or even bounded-below filtration. Convergence of such spectral sequences is a subtle phenomenon: the individual pages $E_r^{p,q}$ may fail to stabilise, the abutment of the tower (its homotopy colimit) need not agree with the object one started with, and even when the pages do stabilise the resulting filtration on $\mathrm{Hom}_{DM}(B, A)$ may fail to be exhaustive, separated, or complete. Boardman's formalism (recalled in §1.4) isolates exactly the two obstructions that must vanish: the map $c : \mathrm{hocolim}_n bc_{\leq n}A \rightarrow A$ from the homotopy colimit of the tower back to A must be an isomorphism (this controls exhaustiveness of the abutment), and the derived limit term $RE_{\infty}^{p,q}$ must vanish (this controls completeness). The bulk of this thesis — Theorem 2.13 and its proof via Corollary 2.12 and Theorem 2.11 — is devoted to establishing both facts for DM , using in an essential way that the source object B is a shifted Tate twist $M(X)(r)[s]$ of the motive of a smooth scheme, so that B is both compact and orthogonal to $DM^{\mathrm{eff}}(n)$ for n large (Proposition 2.8), which forces the spectral sequence into a half-plane and makes the derived limit term vanish automatically (Note 1.55).

We also show that the analogous results fail in the $\mathbb{A}^1$ -stable homotopy category of Morel–Voevodsky $\mathcal{SH}$, providing explicit counterexamples (Proposition 2.16 and Corollary 2.18). The mechanism behind this failure is instructive: the argument that makes $M(X)(r)[s]$ well behaved in DM uses that a motive of a d -dimensional scheme is automatically orthogonal to $DM^{\mathrm{eff}}(n)$ once $n > d + r$ — a rigidity coming from the vanishing of motivic cohomology in negative weights. No such rigidity holds in $\mathcal{SH}$: the unit object $\mathbf{1} \in \mathcal{SH}$ has nonzero slices $s_m \mathbf{1}$ in every degree $m \geq 0$ (Voevodsky, [35]), so $\mathbf{1}$ is never orthogonal to $\mathcal{SH}^{\mathrm{eff}}(n)$ for large n , and the orthogonal tower on $\mathbf{1}$ never stabilises. On the other hand, we verify that, under appropriate conditions on $A \in \mathcal{SH}$ — precisely, that A already lies in some $\mathcal{SH}^{\perp}(r)$ — the spectral sequence is strongly convergent (Proposition 2.19), and that these conditions are satisfied for the Voevodsky slices $s_m G \in \mathcal{SH}$. As a direct consequence, we obtain a spectral sequence strongly convergent at the E_1 term of the Voevodsky slice spectral sequence (Corollary 2.20). Thus the orthogonal spectral sequence is well behaved precisely on the pieces that the slice filtration itself produces, even though it is ill behaved on the ambient category $\mathcal{SH}$.

Outline. Section 1 collects the background material used throughout: triangulated categories, Bousfield localisation and Neeman’s adjoint functor theorem (§1.1–2), the language of exact couples and Boardman’s convergence criteria for spectral sequences (§1.4), and the construction of the triangulated category of mixed motives DM together with Voevodsky’s slice tower (§1.5). Readers already familiar with this material may proceed directly to Section 2, which contains the new results of the thesis: the construction of the orthogonal tower and its associated spectral sequence (§2.1), the proof of strong convergence in DM culminating in Theorem 2.13 (§2.2), and the comparison with the $\mathbb{A}^1$ -stable homotopy category $\mathcal{SH}$, including the counterexample of Proposition 2.16 and the positive result for slices, Corollary 2.20 (§2.3). Section 2.4 surveys three subsequent applications of the orthogonal and slice towers, due to Peláez and collaborators, to the classical theory of adequate equivalence relations on algebraic cycles, and the thesis closes with further results, worked examples, and a discussion of the problems that remain open in §2.4.3.

Note 0.1. The main results of this thesis (Theorem 2.13 and its consequences) were subsequently published as [5].

1 Preliminaries

1.1 Triangulated categories

The main references for this section are [23, 14].

Let $\mathcal{A}$ be an additive category equipped with an autoequivalence $\Sigma : \mathcal{A} \rightarrow \mathcal{A}$ of categories. For every integer $n \in \mathbb{Z}$ and every object X (resp. morphism f) of $\mathcal{A}$, we define $X[n] := \Sigma^n(X)$ (resp. $f[n] = \Sigma^n(f)$).

Definition 1.1 (Verdier). Let $\mathcal{A}$ be an additive category. The structure of a *triangulated category* is given by an autoequivalence $\Sigma = [1] : \mathcal{A} \rightarrow \mathcal{A}$, called the *translation functor*, and a class of diagrams in $\mathcal{A}$. A *triangle* of $(\mathcal{A}, \Sigma)$ is a sequence of morphisms

$$X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1].$$

A morphism between distinguished triangles is a triple (f, g, h) of morphisms of $\mathcal{A}$ making the following diagram commute:

$$\begin{array}{ccccccc} X & \xrightarrow{u} & Y & \xrightarrow{v} & Z & \xrightarrow{w} & X[1] \\ \downarrow f & & \downarrow g & & \downarrow h & & \downarrow f[1] \\ X' & \xrightarrow{u'} & Y' & \xrightarrow{v'} & Z' & \xrightarrow{w'} & X'[1]. \end{array}$$

Such a triple (f, g, h) is an isomorphism if the vertical arrows are isomorphisms. We require the distinguished triangles to satisfy the following axioms:

(TR1) (a) Every triangle of the form $X \xrightarrow{1_X} X \rightarrow 0 \rightarrow X[1]$ is distinguished.

- (b) Every triangle isomorphic to a distinguished triangle is distinguished.
- (c) For every morphism $f : X \rightarrow Y$ in $\mathcal{A}$, there exists a distinguished triangle of the form $X \xrightarrow{f} Y \rightarrow Z \rightarrow X[1]$.
- (TR2) A triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ is distinguished if and only if $Y \xrightarrow{v} Z \xrightarrow{-w} X[1] \xrightarrow{-\Sigma u} Y[1]$ is distinguished.
- (TR3) Suppose we have a commutative diagram of distinguished triangles with vertical arrows f and g . Then there exists (not necessarily unique) an arrow $h : Z \rightarrow Z'$ making the full diagram commute.
- (TR4) **Octahedral axiom.** (We refer to [14] for the precise statement.)

Remark 1.2. A distinguished triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ should be thought of as an abstraction of a cofiber sequence $X \rightarrow Y \rightarrow Y/X$ in topology, or of a short exact sequence $0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$ in an abelian category, with Z playing the role of the “cone” or “cokernel” of u and the connecting map $w : Z \rightarrow X[1]$ recording the failure of u to be an honest monomorphism. Axiom (TR1)(c) guarantees that every morphism has a cone; axiom (TR3) is the analogue of the snake lemma, producing a (non-unique, in general) comparison map on cones; and the octahedral axiom (TR4) governs how cones of composites relate to iterated cones, playing the role played by the third isomorphism theorem for quotients in an abelian category. Throughout this thesis the triangulated category DM (constructed in §1.5) should be kept in mind as the running example: its distinguished triangles come from mapping cones of complexes of sheaves with transfers, and $\mathrm{Hom}_{DM}(-, -)$ plays the role of the derived Hom of sheaf theory.

Note 1.3. If $\mathcal{T} = (\mathcal{A}, \Sigma)$ is a triangulated category, then $\mathcal{T}^{op} = (\mathcal{A}^{op}, \Sigma^{-1})$ is also a triangulated category.

Proposition 1.4. Let $\mathcal{T}$ be a triangulated category. Suppose $\{X_\lambda\}_{\lambda \in \Lambda}$ is a set of objects of $\mathcal{T}$ and that the coproduct $\coprod_{\lambda \in \Lambda} X_\lambda$ exists in $\mathcal{T}$. Then the natural map

$$\coprod_{\lambda \in \Lambda} X_\lambda[1] \longrightarrow \left(\coprod_{\lambda \in \Lambda} X_\lambda \right)[1]$$

is an isomorphism.

Definition 1.5. Let $\mathcal{T}$ be a triangulated category and $\mathcal{A}$ an abelian category. A functor $H : \mathcal{T} \rightarrow \mathcal{A}$ is *homological* if for each distinguished triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ the sequence $H(X) \xrightarrow{H(u)} H(Y) \xrightarrow{H(v)} H(Z) \xrightarrow{H(w)} H(X[1])$ is exact in $\mathcal{A}$.

By axiom (TR2) the sequence extends to a long exact sequence in both directions:

$$\cdots \rightarrow H(Z[-1]) \xrightarrow{H(w[-1])} H(X) \xrightarrow{H(u)} H(Y) \xrightarrow{H(v)} H(Z) \xrightarrow{H(w)} H(X[1]) \rightarrow \cdots$$

Note 1.6. A homological functor on $\mathcal{T}^{op}$ is a *cohomological functor* on $\mathcal{T}$.

Lemma 1.7. Let $\mathcal{T}$ be a triangulated category and U an object of $\mathcal{T}$. The functor $\text{Hom}(U, -) : \mathcal{T} \rightarrow \mathbf{Ab}$ is homological.

Proof. Let $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ be a distinguished triangle. We need exactness at $\text{Hom}(U, Y)$. First, $vu = 0$: by (TR1)(a) and (TR3) applied to the canonical triangle $X \xrightarrow{\text{id}} X \rightarrow 0 \rightarrow X[1]$ and our triangle, we obtain a morphism $h : U \rightarrow X$ (via the map $U \rightarrow 0$) with $h = 0$, and the commutative square forces $vu = 0$. For exactness, if $vf = 0$ then by (TR2) and (TR3) one constructs $h : U \rightarrow X$ with $uh = f$; see [14] for details. $\square$

Note 1.8. It follows that $\text{Hom}(-, U) : \mathcal{T}^{op} \rightarrow \mathbf{Ab}$ is cohomological.

Proposition 1.9. Let $\mathcal{T}$ be a triangulated category and (f, g, h) a morphism of triangles.

1. If two of the three vertical morphisms are isomorphisms, so is the third.
2. If $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{0} X[1]$ is a distinguished triangle, then u is an isomorphism if and only if $Z \cong 0$.
3. Every triangle of the form $X \rightarrow Y \rightarrow Z \xrightarrow{0} X[1]$ is isomorphic to $X \rightarrow X \oplus Z \rightarrow Z \xrightarrow{0} X[1]$.

Proof. See [14], IV §1. $\square$

Definition 1.10. A *tensor structure* on an additive category $\mathcal{A}$ is given by a bifunctor $\otimes : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ which is associative, commutative and unital; that is, there is a unit object 1 and isomorphisms $X \otimes (Y \otimes Z) \xrightarrow{\sim} (X \otimes Y) \otimes Z$, $X \otimes Y \xrightarrow{\sim} Y \otimes X$, $1 \otimes X \xrightarrow{\sim} X$, $X \otimes 1 \xrightarrow{\sim} X$, making the obvious diagrams commute.

Definition 1.11. If $\mathcal{A}$ has a translation $X \mapsto \Sigma X$, the tensor structure is *graded* if

1. $\Sigma \circ (- \otimes -) = \Sigma(-) \otimes -$ and $\Sigma^2(-) \otimes - = - \otimes \Sigma^2(-)$ as functors $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$.
2. The natural isomorphisms $X \otimes \Sigma Y \cong \Sigma X \otimes Y = \Sigma(X \otimes Y) = \Sigma X \otimes Y$ satisfy the condition that the composition $X \otimes \Sigma^2 Y \rightarrow X \otimes \Sigma^2 Y$ is the identity.

Definition 1.12. A *tensor triangulated category* is a triangulated category $\mathcal{T}$ which is also a tensor category (with $\otimes$ graded) such that for each distinguished triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ and every $W \in \mathcal{T}$, the sequence

$$X \otimes W \xrightarrow{u \otimes \text{id}_W} Y \otimes W \xrightarrow{v \otimes \text{id}_W} Z \otimes W \xrightarrow{w \otimes \text{id}_W} X[1] \otimes W = (X \otimes W)[1]$$

is a distinguished triangle.

Definition 1.13. Let $(\mathcal{T}, \Sigma)$ and $(\mathcal{T}', \Sigma')$ be triangulated categories. An additive functor $F : \mathcal{T} \rightarrow \mathcal{T}'$ is *triangulated* if there is a natural isomorphism $F \circ \Sigma \cong \Sigma' \circ F$ such that for every distinguished triangle $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} X[1]$ in $\mathcal{T}$, the triangle $FX \xrightarrow{u} FY \xrightarrow{v} FZ \xrightarrow{w} F(X[1])$ is distinguished in $\mathcal{T}'$.

Definition 1.14. A *triangulated subcategory* of $\mathcal{T}$ is a subcategory $\iota : \mathcal{S} \subset \mathcal{T}$ with the structure of a triangulated category such that the inclusion functor ι is triangulated.

Proposition 1.15. Let $\mathcal{T}$ and $\mathcal{T}'$ be triangulated categories, and consider an adjoint pair of functors where F is the left adjoint:

$$\mathcal{T} \begin{matrix} \xrightarrow{F} \\ \xleftarrow{G} \end{matrix} \mathcal{T}'.$$

Then F is triangulated if and only if G is triangulated.

Proof. See [24], Lem. 5.3.6. □

Proposition 1.16. If $\mathcal{S} \subset \mathcal{T}$ is a full subcategory of a triangulated category $\mathcal{T}$, then $\mathcal{S}$ is a triangulated subcategory if and only if it is invariant under the translation functor and for every distinguished triangle $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ in $\mathcal{T}$ with $X, Y \in \mathcal{S}$, the object Z is isomorphic to an object in $\mathcal{S}$.

1.1.1 Verdier localisation

Definition 1.17. Let $F : \mathcal{D} \rightarrow \mathcal{T}$ be a triangulated functor. The *kernel* of F is the full subcategory $\mathcal{C} = \{X \in \mathcal{D} : F(X) \cong 0\}$.

Proposition 1.18. The kernel of a triangulated functor $F : \mathcal{D} \rightarrow \mathcal{T}$ satisfies:

1. $\mathcal{C}$ is a triangulated subcategory of $\mathcal{D}$.
2. If $X \oplus Y \in \mathcal{C}$, then $X, Y \in \mathcal{C}$.

Proof. See [24], Lem. 2.1.4 and Lem. 2.1.5. □

Definition 1.19. A full triangulated subcategory $\mathcal{C}$ of a triangulated category $\mathcal{D}$ is *thick* if $\mathcal{C}$ is closed under direct summands.

Theorem 1.20 (Verdier [32]). Let $\mathcal{T}$ be a small triangulated category and $\mathcal{C} \subset \mathcal{T}$ a triangulated subcategory. There exists a triangulated category $\mathcal{T}/\mathcal{C}$ and a triangulated functor $\mathcal{Q}_{\mathcal{C}} : \mathcal{T} \rightarrow \mathcal{T}/\mathcal{C}$ such that, for every triangulated functor $F : \mathcal{T} \rightarrow \mathcal{T}'$ with $\mathcal{C}$ contained in the kernel of F , there exists a unique triangulated functor $\bar{F} : \mathcal{T}/\mathcal{C} \rightarrow \mathcal{T}'$ making the diagram commute. The canonical map $\mathcal{Q}_{\mathcal{C}}$ is called Verdier localisation and the category $\mathcal{T}/\mathcal{C}$ is also written $\mathcal{T}[\mathcal{C}^{-1}]$.

1.1.2 Bousfield localisation

Definition 1.21. Let $\mathcal{T}$ be a triangulated category and $\mathcal{S} \subset \mathcal{T}$ a thick subcategory. A *Bousfield localisation functor* exists for $\mathcal{S} \subset \mathcal{T}$ if there exists a right adjoint to the natural functor $F : \mathcal{T} \rightarrow \mathcal{T}/\mathcal{S}$. When this is the case, we call the right adjoint the *Bousfield localisation functor* and denote it $G : \mathcal{T}/\mathcal{S} \rightarrow \mathcal{T}$.

Note 1.22. If $\mathcal{S} \subset \mathcal{T}$ is thick and a Bousfield localisation exists, then by adjunction $\mathrm{Hom}_{\mathcal{T}}(S, GT) = 0$ for every $S \in \mathcal{S}$.

Definition 1.23. An object $T \in \mathcal{T}$ is $\mathcal{S}$ -local if $\mathrm{Hom}_{\mathcal{T}}(S, T) = 0$ for every $S \in \mathcal{S}$.

Proposition 1.24. Let $\mathcal{S}$ be a thick subcategory of a triangulated category $\mathcal{T}$. A Bousfield localisation exists for $\mathcal{S} \subset \mathcal{T}$ if and only if the inclusion $\mathcal{S} \hookrightarrow \mathcal{T}$ admits a right adjoint.

Proof. [24], Prop. 9.1.18. □

Proposition 1.25. Suppose a Bousfield localisation functor $G : \mathcal{T}/\mathcal{S} \rightarrow \mathcal{T}$ exists. For $T \in \mathcal{T}$ there is a distinguished triangle

$$T_{\mathcal{S}} \longrightarrow T \xrightarrow{\eta_T} GT \longrightarrow T_{\mathcal{S}}[1]$$

in which (1) GT is $\mathcal{S}$ -local and (2) $T_{\mathcal{S}} \in \mathcal{S}$.

Remark 1.26. Proposition 1.25 says that every object T decomposes, functorially and up to a distinguished triangle, into an “ $\mathcal{S}$ -torsion part” $T_{\mathcal{S}} \in \mathcal{S}$ and an “ $\mathcal{S}$ -local part” GT that sees nothing of $\mathcal{S}$ (Note 1.22). This is the exact categorical analogue of the decomposition of an abelian group into its torsion subgroup and its torsion-free (localised) quotient, or of the arithmetic fracture square decomposing a spectrum into its rationalisation and its completions at each prime. The orthogonal tower construction of Definition 2.1 below is built from exactly this machinery: at each stage n we take $\mathcal{S} = DM^{\mathrm{eff}}(n+1)$, so that $bc_{\leq n}E = j_{n+1}p_{n+1}E$ is the “ $DM^{\mathrm{eff}}(n+1)^{\perp}$ -local part” of E , i.e. the part of E that is orthogonal to (does not see maps from) motives of Tate twist $\geq n+1$. Increasing n enlarges the class of motives being tested against, so $bc_{\leq n}E$ should be thought of as an increasingly fine approximation to E , in analogy with a Postnikov tower approximating a spectrum by killing higher and higher homotopy.

1.2 Neeman’s adjoint functor theorem

Definition 1.27. Let $\mathcal{T}$ be a triangulated category admitting coproducts.

1. An object $U \in \mathcal{T}$ is *compact* if $\mathrm{Hom}_{\mathcal{T}}(U, \coprod_{\lambda} X_{\lambda}) = \coprod_{\lambda} \mathrm{Hom}_{\mathcal{T}}(U, X_{\lambda})$ for every coproduct of objects in $\mathcal{T}$.
2. $\mathcal{T}$ is *compactly generated* if there exists a set $\mathcal{U}$ of compact objects such that if $\mathrm{Hom}_{\mathcal{T}}(U, X) = 0$ for every $U \in \mathcal{U}$, then $X = 0$.

Definition 1.28. Let $\mathcal{T}$ be a triangulated category admitting coproducts and $X_0 \rightarrow X_1 \rightarrow X_2 \rightarrow \cdots$ a sequence in $\mathcal{T}$. The *homotopy colimit* $\text{hocolim } X_i$ is defined (up to non-canonical isomorphism) by the distinguished triangle

$$\coprod_i X_i \xrightarrow{\text{id}-s} \coprod_i X_i \longrightarrow \text{hocolim } X_i \longrightarrow \left(\coprod_i X_i \right)[1],$$

where $s : X_{i_0} \rightarrow X_{i_0+1} \rightarrow \coprod_i X_i$ is the composition into the factor X_{i_0+1} .

Proposition 1.29. Let U be a compact object in a triangulated category $\mathcal{T}$ admitting coproducts. If $X_0 \rightarrow X_1 \rightarrow \cdots$ is a sequence in $\mathcal{T}$, then

$$\text{Hom}_{\mathcal{T}}(U, \text{hocolim } X_i) = \text{colim } \text{Hom}_{\mathcal{T}}(U, X_i).$$

Proof. [23], Lem. 2.8. □

Theorem 1.30 (Brown representability [23]). Let $\mathcal{T}$ be a compactly generated triangulated category and $H : \mathcal{T}^{\text{op}} \rightarrow \mathbf{Ab}$ a cohomological functor sending coproducts to products. Then H is representable.

Theorem 1.31 (Neeman's adjoint functor theorem [23], Thm. 4.1). Let $\mathcal{S}$ be a compactly generated triangulated category and $\mathcal{T}$ any triangulated category. Let $F : \mathcal{S} \rightarrow \mathcal{T}$ be a triangulated functor respecting coproducts. Then F has a right adjoint $G : \mathcal{T} \rightarrow \mathcal{S}$.

1.3 The derived category and derived functors

Definition 1.32. Let $\mathcal{A}$ be an additive category. A *complex* $A^\bullet$ in $\mathcal{A}$ is a diagram

$$A^\bullet : \quad \cdots \longrightarrow A^{n-1} \xrightarrow{d^{n-1}} A^n \xrightarrow{d^n} A^{n+1} \xrightarrow{d^{n+1}} \cdots$$

with $d^n \circ d^{n-1} = 0$ for all $n \in \mathbb{Z}$. It is *bounded* if $A^n = 0$ for $|n| \gg 0$; *bounded below* if $A^n = 0$ for $n \ll 0$; *bounded above* if $A^n = 0$ for $n \gg 0$.

Definition 1.33. Let $\mathcal{A}$ be an abelian category. Setting $\mathcal{S} = \{A^\bullet : H^n(A^\bullet) = 0 \text{ for all } n \in \mathbb{Z}\}$ (the acyclic complexes), $\text{Mor}_{\mathcal{S}}$ consists of all quasi-isomorphisms and the *derived category* of $\mathcal{A}$ is

$$D(\mathcal{A}) = K(\mathcal{A})[\mathcal{S}^{-1}]$$

in the sense of Theorem 1.20.

Proposition 1.34. Let $\mathcal{A}$ be an abelian category with enough injectives. For $X, Y \in \mathcal{A}$,

$$\text{Hom}_{D(\mathcal{A})}(X, Y[i]) = \begin{cases} \text{Ext}_{\mathcal{A}}^i(X, Y) & i \geq 0, \\ 0 & i < 0. \end{cases}$$

1.4 Spectral sequences

We follow Boardman's article [13] closely and supplement it with complete proofs and diagrams. The key novelty with respect to classical accounts is the systematic use of the limit groups D^∞ and $D^{-\infty}$ as the natural targets for convergence.

1.4.1 Basic definitions

Definition 1.35. A *(bigraded cohomological) spectral sequence* is a collection of data:

1. abelian groups E_r^{pq} for all $r \geq 1$ and $p, q \in \mathbb{Z}$;
2. morphisms $d_r^{pq} : E_r^{pq} \rightarrow E_r^{p+r, q-r+1}$ with $(d_r)^2 = 0$;
3. isomorphisms $E_{r+1}^{pq} \cong H^{pq}(E_r^{**}, d_r) := \ker d_r^{pq} / \operatorname{im} d_r^{p-r, q+r-1}$.

The integer $n = p + q$ is the *total degree*, p the *filtration degree*, and q the *complementary degree*.

For each fixed (p, q) we have a nested sequence of subgroups of E_1^{pq} :

$$0 = B_1^{pq} \subset B_2^{pq} \subset \cdots \subset B_r^{pq} \subset \cdots \subset Z_r^{pq} \subset \cdots \subset Z_2^{pq} \subset Z_1^{pq} = E_1^{pq},$$

where $Z_r^{pq} = \ker(d_r^{pq} : E_r^{pq} \rightarrow E_r^{p+r, q-r+1})$ and $B_r^{pq} = \operatorname{im}(d_r^{p-r, q+r-1} : E_r^{p-r, q+r-1} \rightarrow E_r^{pq})$ are the *cycles* and *boundaries* at stage r , viewed as subgroups of E_1^{pq} via the successive isomorphisms.

Remark 1.36. Concretely, $E_{r+1}^{pq} = Z_r^{pq} / B_r^{pq}$ is obtained from E_r^{pq} by discarding the elements killed by d_r going out (these are not “cycles”) and further quotienting by the elements hit by d_r coming in (these are “boundaries”). As r grows, both the differentials d_r and their sources and targets move further apart in bidegree, so in any *fixed* bidegree (p, q) the groups E_r^{pq} eventually stabilise once r is larger than the maximum of the distances to the nearest possibly-nonzero neighbouring group — this is the case, for instance, whenever the spectral sequence is confined to a half-plane (Note 1.55). The subtlety addressed by Definition 1.37 and the convergence criteria of §1.4.2–§1.4.5 is that “eventually stable in each bidegree” is *not* automatically the same as “correctly computes the associated graded of the abutment”: the two can differ by the derived limit term RE_∞^{pq} of Lemma 1.48, which is why strong convergence requires an extra hypothesis beyond mere degree-wise stabilisation.

Definition 1.37. The *infinite cycles* and *infinite boundaries* are

$$Z_\infty^{pq} = \bigcap_{r \geq 1} Z_r^{pq}, \quad B_\infty^{pq} = \bigcup_{r \geq 1} B_r^{pq},$$

and the E_∞ -term is the sub-quotient $E_\infty^{pq} = Z_\infty^{pq} / B_\infty^{pq}$.

Definition 1.38. Let H be an abelian group. A *decreasing filtration* of H is a sequence of subgroups

$$\dots \subset F^{p+1}H \subset F^pH \subset \dots \subset H, \quad p \in \mathbb{Z}.$$

Write $F^\infty H := \bigcap_p F^pH$ and $F^{-\infty}H := \bigcup_p F^pH$. The filtration is:

- *exhaustive* if $F^{-\infty}H = H$;
- *separated* (or Hausdorff) if $F^\infty H = 0$;
- *complete* if $R^1\varprojlim_p F^pH = 0$, equivalently $H \xrightarrow{\sim} \varprojlim_p H/F^pH$.

The p -th *graded piece* is $\mathrm{Gr}_F^p H = F^pH/F^{p+1}H$.

Definition 1.39. Let E_r^{pq} be a spectral sequence and (H^*, F^*) a filtered graded abelian group. The spectral sequence *abuts to* (or *converges to*) (H^*, F^*) if there are isomorphisms

$$E_\infty^{pq} \xrightarrow{\sim} \mathrm{Gr}_F^p H^{p+q}$$

for every (p, q) . One writes $E_1^{pq} \Rightarrow H^n$ (where $n = p + q$).

- (i) The sequence *converges weakly* to (H^*, F^*) if additionally F^* is exhaustive.
- (ii) It *converges* if F^* is exhaustive and separated.
- (iii) It *converges strongly* if F^* is exhaustive, separated, and complete.

Strong convergence allows one to recover H^n , up to extensions, from the E_∞ -terms via the filtration.

Definition 1.40. A *morphism of spectral sequences* $f_r^{pq} : E_r^{pq} \rightarrow \tilde{E}_r^{pq}$ is a collection of group homomorphisms, one for each $r \geq 1$ and (p, q) , satisfying:

1. f_r^{pq} commutes with differentials: $\tilde{d}_r^{pq} \circ f_r^{pq} = f_r^{p+r, q-r+1} \circ d_r^{pq}$, i.e. the square

$$\begin{array}{ccc} E_r^{pq} & \xrightarrow{d_r^{pq}} & E_r^{p+r, q-r+1} \\ f_r^{pq} \downarrow & & \downarrow f_r^{p+r, q-r+1} \\ \tilde{E}_r^{pq} & \xrightarrow{\tilde{d}_r^{pq}} & \tilde{E}_r^{p+r, q-r+1} \end{array}$$

commutes for all r, p, q .

2. The induced map on cohomology agrees with f_{r+1}^{pq} under the structure isomorphisms $E_{r+1}^{pq} \cong H(E_r^{**}, d_r)$.

Theorem 1.41 (Comparison theorem for strongly convergent spectral sequences). *Let $E_r^{pq} \Rightarrow H^n$ and $\tilde{E}_r^{pq} \Rightarrow \tilde{H}^n$ be two spectral sequences converging strongly to the filtered groups (H^*, F^*) and $(\tilde{H}^*, \tilde{F}^*)$ respectively. Let $f_r : E_r^{**} \rightarrow \tilde{E}_r^{**}$ be a*

morphism of spectral sequences and $f^* : H^* \rightarrow \tilde{H}^*$ a morphism of filtered graded groups such that the diagram

$$\begin{array}{ccc} E_\infty^{pq} & \xrightarrow{\sim} & \mathrm{Gr}_F^p H^{p+q} \\ f_\infty^{pq} \downarrow & & \downarrow \mathrm{Gr}^p f^{p+q} \\ \tilde{E}_\infty^{pq} & \xrightarrow{\sim} & \mathrm{Gr}_{\tilde{F}}^p \tilde{H}^{p+q} \end{array}$$

commutes for all (p, q) . If there exists $a \geq 1$ such that $f_a^{pq} : E_a^{pq} \xrightarrow{\sim} \tilde{E}_a^{pq}$ is an isomorphism for all (p, q) , then f_a^{pq} is an isomorphism for all $r \geq a$ (in particular at $r = \infty$), and $f^n : H^n \rightarrow \tilde{H}^n$ is an isomorphism of filtered groups for all n .

Proof. Since f_a^{pq} is an isomorphism and commutes with d_a , the induced map on cohomology $f_{a+1}^{pq} : E_{a+1}^{pq} \rightarrow \tilde{E}_{a+1}^{pq}$ is also an isomorphism (the five lemma applied to the short exact sequence $0 \rightarrow B_a^{pq} \rightarrow Z_a^{pq} \rightarrow E_{a+1}^{pq} \rightarrow 0$). By induction, f_r^{pq} is an isomorphism for all $r \geq a$, hence at $r = \infty$.

The commutative diagram in the statement gives, for each p and $n = p + q$, an isomorphism $\mathrm{Gr}_F^p H^n \xrightarrow{\sim} \mathrm{Gr}_{\tilde{F}}^p \tilde{H}^n$. Since both filtrations are exhaustive, separated, and complete, we apply the following comparison lemma: a morphism of complete, separated, exhaustive filtrations on H^n and $\tilde{H}^n$ that induces isomorphisms on all graded pieces is itself an isomorphism of filtered groups. This follows by induction on the filtration length (for finite filtrations) and in general by passing to the limit using completeness:

$$H^n \cong \varprojlim_p H^n / F^p H^n, \quad \tilde{H}^n \cong \varprojlim_p \tilde{H}^n / \tilde{F}^p \tilde{H}^n,$$

and the induced maps on each finite quotient $H^n / F^p H^n \rightarrow \tilde{H}^n / \tilde{F}^p \tilde{H}^n$ are isomorphisms (by the five lemma applied to the finite filtration $F^p \supset F^{p+1} \supset \dots$), so the inverse limit is an isomorphism. $\square$

1.4.2 Spectral sequences from exact couples

Definition 1.42. An *exact couple* is a diagram of abelian groups and morphisms

$$\begin{array}{ccc} D & \xrightarrow{\alpha} & D \\ & \swarrow \gamma \quad \searrow \beta & \\ & E & \end{array}$$

that is exact at each of the three vertices, i.e. $\ker \alpha = \mathrm{im} \gamma$, $\ker \beta = \mathrm{im} \alpha$, and $\ker \gamma = \mathrm{im} \beta$.

The composite $d := \beta \circ \gamma : E \rightarrow E$ satisfies $d^2 = \beta \gamma \beta \gamma = 0$ (since $\gamma \beta = 0$ by exactness at D). Thus (E, d) is a differential group.

Proposition 1.43. *Given an exact couple $(D, E, \alpha, \beta, \gamma)$, set*

$$D' := \operatorname{im} \alpha \subset D, \quad E' := H(E, d) = \ker d / \operatorname{im} d.$$

Define morphisms $\alpha' : D' \rightarrow D'$, $\beta' : D' \rightarrow E'$, $\gamma' : E' \rightarrow D'$ by

$$\alpha'(\alpha x) := \alpha(\alpha x), \quad \beta'(\alpha x) := [\beta x], \quad \gamma'([e]) := \gamma(e),$$

where $[-]$ denotes the class in E' . Then:

1. α', β', γ' are well-defined.
2. The derived couple $(D', E', \alpha', \beta', \gamma')$

$$\begin{array}{ccc} D' & \xrightarrow{\alpha'} & D' \\ & \nwarrow \gamma' \quad \nearrow \beta' & \\ & E' & \end{array}$$

is again an exact couple.

Iterating, one obtains a sequence of exact couples $(D_r, E_r, \alpha_r, \beta_r, \gamma_r)$ with $D_1 = D$, $E_1 = E$, and $d_r := \beta_r \circ \gamma_r : E_r \rightarrow E_r$. The groups $\{E_r, d_r\}$ form a spectral sequence.

Proof. Well-definedness of β' . Suppose $\alpha x = \alpha x'$, so $\alpha(x - x') = 0$, hence $x - x' \in \ker \alpha = \operatorname{im} \gamma$, say $x - x' = \gamma e$. Then $\beta x - \beta x' = \beta \gamma e = d e$, so $[\beta x] = [\beta x']$ in E' .

Well-definedness of γ' . Suppose $[e] = [e']$, so $e - e' \in \operatorname{im} d = \operatorname{im} \beta \gamma$. Write $e - e' = \beta \gamma f$ for some $f \in E$. Then $\gamma(e) - \gamma(e') = \gamma \beta \gamma f$. But $\gamma \beta = 0$ by exactness, so $\gamma(e) = \gamma(e')$ and γ' is well-defined. Moreover $\gamma'[e] = \gamma e \in \ker \alpha = \operatorname{im} \gamma \subset D'$, so γ' maps into D' .

Exactness at D' (i.e. $\ker \alpha' = \operatorname{im} \gamma'$). $\ker \alpha' = \ker(\alpha|_{\operatorname{im} \alpha}) = \operatorname{im} \alpha \cap \ker \alpha = \operatorname{im} \alpha \cap \operatorname{im} \gamma$. On the other hand $\operatorname{im} \gamma' = \gamma(\ker d) = \gamma(\ker \beta \gamma)$. If $\gamma e \in \operatorname{im} \alpha$ then $\gamma e = \alpha x$ and $\beta \gamma e = \beta \alpha x = 0$ (exactness at D), so $e \in \ker \beta \gamma = \ker d$. Conversely if $e \in \ker d$ then $\beta \gamma e = 0$, so $\gamma e \in \ker \beta = \operatorname{im} \alpha$. Hence $\ker \alpha' = \operatorname{im} \gamma'$.

Exactness at D' on the other side (i.e. $\ker \beta' = \operatorname{im} \alpha'$). $\ker \beta' = \{\alpha x : [\beta x] = 0\} = \{\alpha x : \beta x \in \operatorname{im} d\} = \{\alpha x : \beta x = \beta \gamma e \text{ for some } e\}$. Since $\beta(x - \gamma e) = 0$, we have $x - \gamma e \in \ker \beta = \operatorname{im} \alpha$, say $x - \gamma e = \alpha y$. Then $\alpha x = \alpha^2 y = \alpha'(\alpha y)$. Hence $\ker \beta' \subset \operatorname{im} \alpha'$; the reverse inclusion is clear.

Exactness at E' . $\ker \gamma' = \{[e] \in E' : \gamma e = 0\} = \{[e] : e \in \ker \gamma\} = \{[e] : e \in \operatorname{im} \beta\}$. On the other hand $\operatorname{im} \beta' = \{[\beta x] : x \in D\} = \beta(\ker \beta \gamma^{-1}(\operatorname{im} \alpha)) / \operatorname{im} d$. Since $\beta \alpha = 0$ the image of β lands in $\ker \gamma / \operatorname{im} d$ (by exactness at E , $\ker \gamma = \operatorname{im} \beta$), so $\operatorname{im} \beta' = \{[\beta x]\} \subset \ker \gamma'$ and one checks the reverse inclusion similarly.

Iteration is straightforward: $(D_r, E_r) := (D'_{r-1}, E'_{r-1})$ and $E_{r+1}^{pq} = \ker d_r^{pq} / \operatorname{im} d_r^{p-r, q+r-1}$ recovers the structure isomorphism of Definition 1.35. $\square$

Note 1.44. In the bigraded case, one indexes the exact couple as $(D^{**}, E^{**}, \alpha, \beta, \gamma)$ with

$$\deg \alpha = (-1, 1), \quad \deg \beta = (0, 0), \quad \deg \gamma = (1, 0).$$

After deriving $r - 1$ times, the r -th couple has degrees

$$\deg \alpha_r = (-1, 1), \quad \deg \beta_r = (r - 1, 1 - r), \quad \deg \gamma_r = (1, 0),$$

and the differential $d_r = \beta_r \gamma_r$ has bidegree $(r, 1 - r)$. In the homological convention E_{pq} the differential goes $d_r : E_r^{pq} \rightarrow E_r^{p-r, q+r-1}$.

1.4.3 Convergence and limit groups

Fix a bigraded exact couple $(D^{**}, E^{**}, \alpha, \beta, \gamma)$ as in Note 1.44. Write $\alpha : D^{p,*} \rightarrow D^{p-1,*}$ for the component of α in filtration degree.

Definition 1.45. Define the *colimit group* and *limit group*:

$$D^{-\infty} := \varinjlim_p D^{p,*} \quad (\text{under } \alpha), \quad D^{\infty} := \varprojlim_p D^{p,*} \quad (\text{under } \alpha).$$

Let $\eta^p : D^{p,*} \rightarrow D^{-\infty}$ and $\pi^p : D^{\infty} \rightarrow D^{p,*}$ denote the canonical maps. Define filtrations

$$F^p D^{-\infty} := \text{im } \eta^p, \quad F^p D^{\infty} := \ker \pi^p.$$

Proposition 1.46. *With the notation of Definition 1.45:*

1. *The filtration $F^* D^{-\infty}$ is exhaustive.*
2. *The filtration $F^* D^{\infty}$ is separated and complete.*
3. $\bigcup_p F^p D^{\infty} = \ker(D^{\infty} \rightarrow D^{-\infty})$.

Proof. (1) follows immediately since $D^{-\infty} = \varinjlim_p D^{p,*} = \bigcup_p \text{im } \eta^p = \bigcup_p F^p D^{-\infty}$.

(2) *Separated:* $F^{\infty} D^{\infty} = \bigcap_p \ker \pi^p$. An element $x \in D^{\infty}$ lies in $\bigcap_p \ker \pi^p$ if and only if $\pi^p(x) = 0$ for all p , i.e. $x = 0$ in the limit, so $F^{\infty} D^{\infty} = 0$.

Complete: We must show $R^1 \varprojlim_p F^p D^{\infty} = 0$, i.e. $R^1 \varprojlim_p \ker \pi^p = 0$. For each p there is a short exact sequence

$$0 \longrightarrow \ker \pi^p \longrightarrow D^{\infty} \xrightarrow{\pi^p} D^{p,*} \longrightarrow \text{coker } \pi^p \longrightarrow 0.$$

Taking the inverse limit and using the Milnor sequence for $R^1 \varprojlim$, one checks that $R^1 \varprojlim_p D^{\infty} = 0$ (the system $D^{\infty} \rightrightarrows D^{\infty} \rightrightarrows \dots$ is Mittag-Leffler), which gives the result.

(3) An element $x \in D^{\infty}$ satisfies $x \in F^p D^{\infty} = \ker \pi^p$ for some p if and only if $\pi^p(x) = 0 \in D^{p,*}$, i.e. the image of x under $D^{\infty} \rightarrow D^{p,*} \rightarrow D^{-\infty}$ is zero. Since the colimit is the union of the images, x maps to 0 in $D^{-\infty}$ if and only if it maps to 0 in $D^{p,*}$ for some p (as every element of $D^{-\infty}$ comes from some finite stage), which is precisely $\bigcup_p \ker \pi^p$. $\square$

Remark 1.47. There is a short exact sequence of abelian groups

$$0 \longrightarrow R^1 \varprojlim_p D^{p+1,*} \longrightarrow D^\infty \longrightarrow \varprojlim_p D^{p,*} / D^{p+1,*} \longrightarrow 0,$$

obtained from the Milnor exact sequence applied to the tower $\cdots \rightarrow D^{p+1,*} \rightarrow D^{p,*} \rightarrow \cdots$. The term $R^1 \varprojlim$ is the obstruction to strong convergence.

The following key lemma describes the relationship between E_∞ and the filtration on the limit groups. It is the technical heart of Boardman's theory.

Lemma 1.48. *There are natural short exact sequences*

$$0 \longrightarrow \operatorname{im} \beta / B_\infty^{pq} \longrightarrow E_\infty^{pq} \longrightarrow Z_\infty^{pq} / \operatorname{im} \beta \longrightarrow 0, \quad (2)$$

$$0 \longrightarrow \operatorname{Gr}_F^p D^{-\infty} \longrightarrow E_\infty^{pq} \longrightarrow RE_\infty^{pq} \longrightarrow 0, \quad (3)$$

where $RE_\infty^{pq} := R^1 \varprojlim_r Z_r^{pq} / B_r^{pq}$ is the derived E_∞ -term. In particular, if $RE_\infty^{pq} = 0$, then $E_\infty^{pq} \cong \operatorname{Gr}_F^p D^{-\infty}$.

Proof. The Milnor exact sequence applied to the tower $\cdots \supset Z_r^{pq} \supset Z_{r+1}^{pq} \supset \cdots$ gives

$$0 \rightarrow \varprojlim_r Z_r^{pq} \rightarrow Z_1^{pq} \xrightarrow{1-s} Z_1^{pq} \rightarrow R^1 \varprojlim_r Z_r^{pq} \rightarrow 0,$$

where s is the shift. One identifies $\varprojlim_r Z_r^{pq} = Z_\infty^{pq}$ and similarly $\varprojlim_r B_r^{pq} = B_\infty^{pq}$. The quotient short exact sequence then yields (2). Sequence (3) follows from the identification of $\operatorname{Gr}_F^p D^{-\infty}$ with a quotient of Z_∞^{pq} via the map γ ; see [13], §5. $\square$

Definition 1.49. Let E_r^{pq} be the spectral sequence associated to an exact couple $(D^{**}, E^{**}, \alpha, \beta, \gamma)$.

1. The spectral sequence is *conditionally convergent to D^∞* if $D^{-\infty} = 0$.
2. The spectral sequence is *conditionally convergent to $D^{-\infty}$* if $D^\infty = 0$ and $RD^\infty = 0$, where $RD^\infty := R^1 \varprojlim_p D^{p,*}$.

Proposition 1.50. *Suppose E_r^{pq} is associated to an exact couple.*

1. *The filtration $F^* D^{-\infty}$ is always exhaustive (Proposition 1.46(1)).*
2. *If the spectral sequence converges conditionally to D^∞ (i.e. $D^{-\infty} = 0$), then the filtration $F^* D^\infty$ is exhaustive, separated, and complete (Proposition 1.46(2)).*
3. *In either case the spectral sequence abuts weakly: $E_\infty^{pq} \rightarrow \operatorname{Gr}_F^p D^{\pm\infty}$ has kernel and cokernel controlled by RE_∞^{pq} and RD^∞ via the exact sequences of Lemma 1.48.*

Proof. Part (1) is immediate from the definition of $D^{-\infty}$ as a colimit. For part (2), when $D^{-\infty} = 0$ the exact sequence $0 \rightarrow \varprojlim_p F^p D^\infty \rightarrow D^\infty \rightarrow D^{-\infty} = 0$ shows $\varprojlim_p F^p D^\infty = D^\infty$, i.e. the filtration is exhaustive after reindexing; separatedness and completeness are Proposition 1.46(2). Part (3) is a restatement of Lemma 1.48. $\square$

Theorem 1.51 (Boardman [13], Thm. 7.2 and 7.3). *Let E_r^{pq} be the spectral sequence of an exact couple.*

1. (Strong convergence.) *If E_r^{pq} converges conditionally to D^∞ and $RE_\infty^{pq} = 0$ for all (p, q) , then E_r^{pq} converges strongly to D^∞ , i.e. there are isomorphisms $E_\infty^{pq} \xrightarrow{\sim} \text{Gr}_F^p D^\infty$ and the filtration $F^* D^\infty$ is exhaustive, separated, and complete.*
2. (Weak convergence.) *If $D^\infty = 0$, then E_r^{pq} converges weakly to $D^{-\infty}$.*

Proof. Part (1). Since $D^{-\infty} = 0$, by Proposition 1.46 the filtration $F^* D^\infty$ is separated and complete, and the sequence $0 \rightarrow D^\infty \rightarrow \prod_p D^{p,*} \xrightarrow{\alpha} \prod_p D^{p,*} \rightarrow 0$ is exact (Milnor). Consider the natural map from the exact couple: for each p and each r , there is a commutative diagram

$$\begin{array}{ccccccc} D^{p+r,*} & \xrightarrow{\alpha^r} & D^{p,*} & \xrightarrow{\beta} & E^{p,*} & \xrightarrow{\gamma} & D^{p-1,*} \\ \downarrow & & \downarrow \eta^p & & \downarrow \phi_r & & \downarrow \\ D^{p+r,*} & \xrightarrow{\alpha^r} & D^{p,*} & \xrightarrow{\beta} & E_r^{p,*} & \xrightarrow{\gamma_r} & D_r^{p-1,*} \end{array}$$

where $E_r^{p,*} = \text{im}(\alpha^{r-1} : D^{p+r-1,*} \rightarrow D^{p,*}) / \text{im}(\alpha^r : D^{p+r,*} \rightarrow D^{p,*})$ and the dashed vertical arrow is the quotient map. Passing to the limit $r \rightarrow \infty$ yields a map

$$\phi : \text{Gr}_F^p D^\infty \longrightarrow E_\infty^{pq},$$

which is an isomorphism when $RE_\infty^{pq} = 0$ by the exact sequence $0 \rightarrow \text{Gr}_F^p D^{-\infty} \rightarrow E_\infty^{pq} \rightarrow RE_\infty^{pq} \rightarrow 0$ of Lemma 1.48 together with $D^{-\infty} = 0$ (so $\text{Gr}_F^p D^{-\infty} = \text{Gr}_F^p D^\infty$).

The filtration $F^* D^\infty$ is exhaustive because: for any $x \in D^\infty$, x maps to 0 in $D^{p,*}$ for all sufficiently small p (since $D^{-\infty} = \varinjlim_p D^{p,*} = 0$ means every element of each $D^{p,*}$ eventually maps to 0 under α), hence $x \in F^p D^\infty$ for some p .

Part (2). The filtration $F^* D^{-\infty}$ is always exhaustive. When $D^\infty = 0$, separatedness is automatic: $\bigcap_p F^p D^{-\infty} = \bigcap_p \text{im } \eta^p$. An element x in this intersection means $x = \eta^p(x_p)$ for all p , with compatible choices; but the compatibility and $D^\infty = 0$ force $x = 0$. The abutment $E_\infty^{pq} \rightarrow \text{Gr}_F^p D^{-\infty}$ now follows from Lemma 1.48 with RE_∞^{pq} potentially nonzero. $\square$

1.4.4 The spectral sequence associated to a filtered complex

The most classical source of spectral sequences is a filtered complex.

Definition 1.52. Let $\mathcal{A}$ be an abelian category and $K^\bullet$ a cochain complex. A *decreasing filtration* on $K^\bullet$ is a collection of subcomplexes

$$\dots \supset F^{p+1}K^\bullet \supset F^pK^\bullet \supset \dots$$

of $K^\bullet$ (each F^pK^n is a subgroup of K^n , and the differential $d : K^n \rightarrow K^{n+1}$ restricts to $d : F^pK^n \rightarrow F^pK^{n+1}$).

Proposition 1.53. Let $(K^\bullet, F^\bullet)$ be a filtered cochain complex. There is a spectral sequence with

$$E_0^{pq} = F^pK^{p+q}/F^{p+1}K^{p+q}, \quad E_1^{pq} = H^{p+q}(F^pK^\bullet/F^{p+1}K^\bullet),$$

and $d_r : E_r^{pq} \rightarrow E_r^{p+r, q-r+1}$ induced by the differential of $K^\bullet$. The abutment is $H^*(K^\bullet)$ with the filtration $F^pH^n(K^\bullet) := \text{im}(H^n(F^pK^\bullet) \rightarrow H^n(K^\bullet))$.

Proof. This spectral sequence arises from the exact couple

$$\begin{array}{ccc} \bigoplus_{p,q} H^{p+q}(F^pK^\bullet) & \xrightarrow{\alpha} & \bigoplus_{p,q} H^{p+q}(F^pK^\bullet) \\ & \nwarrow \gamma \quad \nearrow \beta & \\ & \bigoplus_{p,q} H^{p+q}(F^pK^\bullet/F^{p+1}K^\bullet) & \end{array}$$

where α is induced by the inclusions $F^{p+1}K^\bullet \hookrightarrow F^pK^\bullet$ (shifting the filtration degree by -1), β is the map on cohomology induced by the quotient $F^pK^\bullet \rightarrow F^pK^\bullet/F^{p+1}K^\bullet$, and γ is the connecting homomorphism from the long exact sequence of the short exact sequence

$$0 \longrightarrow F^{p+1}K^\bullet \longrightarrow F^pK^\bullet \longrightarrow F^pK^\bullet/F^{p+1}K^\bullet \longrightarrow 0.$$

Exactness of the couple at each vertex is exactness of this long exact sequence. One checks that $D^{-\infty} = \varinjlim_p H^n(F^pK^\bullet) = H^n(K^\bullet)$ and $D^\infty = \varprojlim_p H^n(F^pK^\bullet)$, giving the stated abutment. $\square$

1.4.5 Proposition 1.63: the spectral sequence of a tower

The spectral sequence that plays the central role in this thesis arises not from a filtered complex but from a *tower of objects* in a triangulated category, via an exact couple built from the long exact sequences of the tower triangles.

Fix a triangulated category $\mathcal{T}$ admitting coproducts, objects $A, B \in \mathcal{T}$, and a tower

$$\dots \longrightarrow A_{n+1} \xrightarrow{f_n} A_n \longrightarrow \dots \longrightarrow A_1 \xrightarrow{f_0} A_0 \quad (4)$$

of objects and morphisms in $\mathcal{T}$. For each n complete f_n to a distinguished triangle

$$A_{n+1} \xrightarrow{f_n} A_n \longrightarrow Q_n \longrightarrow A_{n+1}[1],$$

so Q_n is the n -th associated graded object.

Proposition 1.54. *In the above setup, application of $\mathrm{Hom}_{\mathcal{T}}(B, -)$ to the tower (4) yields an exact couple*

$$\begin{array}{ccc} \bigoplus_{n,q} \mathrm{Hom}(B, A_n[q]) & \xrightarrow{(f_n)_*} & \bigoplus_{n,q} \mathrm{Hom}(B, A_n[q]) \\ & \swarrow \gamma \quad \searrow \beta & \\ & \bigoplus_{n,q} \mathrm{Hom}(B, Q_n[q]) & \end{array}$$

where $(f_n)_*$ is induced by f_n , β is induced by $A_n \rightarrow Q_n$, and γ is the connecting map $Q_n \rightarrow A_{n+1}[1]$. The associated spectral sequence has E_1 -term

$$E_1^{p,q} = \mathrm{Hom}_{\mathcal{T}}(B, Q_{-p}[q+p])$$

and converges to $\mathrm{Hom}_{\mathcal{T}}(B, \mathrm{hocolim}_n A_n)$ when B is compact.

Proof. For each n , applying the cohomological functor $\mathrm{Hom}_{\mathcal{T}}(B, -)$ (Lemma 1.7) to the distinguished triangle $A_{n+1} \rightarrow A_n \rightarrow Q_n \rightarrow A_{n+1}[1]$ gives a long exact sequence

$$\cdots \rightarrow \mathrm{Hom}(B, A_{n+1}[q]) \xrightarrow{(f_n)_*} \mathrm{Hom}(B, A_n[q]) \xrightarrow{\beta} \mathrm{Hom}(B, Q_n[q]) \xrightarrow{\gamma} \mathrm{Hom}(B, A_{n+1}[q+1]) \rightarrow \cdots$$

Assembling these into the bigraded groups

$$D^{pq} := \mathrm{Hom}(B, A_{-p}[q+p]), \quad E^{pq} := \mathrm{Hom}(B, Q_{-p}[q+p]),$$

the maps $\alpha := (f_n)_*$, β , γ defined entry-wise give the exact couple, since the long exact sequence provides exactness at each vertex by definition. The bidegrees are $\deg \alpha = (-1, 1)$, $\deg \beta = (0, 0)$, $\deg \gamma = (1, 0)$, matching Note 1.44, so the differential $d_r = \beta_r \gamma_r$ has bidegree $(r, 1 - r)$.

For abutment: when B is compact, by Proposition 1.29,

$$\mathrm{Hom}_{\mathcal{T}}(B, \mathrm{hocolim}_n A_n) = \mathrm{colim}_n \mathrm{Hom}_{\mathcal{T}}(B, A_n) = D^{-\infty,*},$$

where the colimit is taken with respect to the tower maps $(f_n)_*$. The filtration on $D^{-\infty} = \mathrm{Hom}(B, \mathrm{hocolim}_n A_n)$ is then exactly the filtration $F_{\bullet}$ from Definition 2.5, and the spectral sequence abuts to $\mathrm{Hom}(B, \mathrm{hocolim}_n A_n)$ by Proposition 1.50. $\square$

Note 1.55. If additionally $E_1^{pq} = 0$ for all p below some bound p_0 (which occurs in our setting because $B \in DM^{\mathrm{eff}}(s)$ forces $\mathrm{Hom}(B, bc_{\leq n} A) = 0$ for $n \leq s - 1$), then the spectral sequence is confined to a half-plane and the hypothesis $RE_{\infty}^{pq} = 0$ of Theorem 1.51(1) is automatically satisfied. This is the key vanishing used in the proof of Theorem 2.13.

1.5 The Voevodsky category of mixed motives

Let k be a field. All schemes are assumed to be separated of finite type over k . We write Sch/k for the category of schemes over k and Sm/k for the full subcategory of smooth schemes.

1.5.1 Finite correspondences

The canonical reference for this subsection is [20], Ch. 1.

Definition 1.56. Let X be a smooth connected scheme over k and Y any scheme over k . An *elementary correspondence* from X to Y is an irreducible closed subscheme $W \subset X \times Y$ which is finite and surjective over X .

The group $\mathrm{Cor}(X, Y)$ is the free abelian group generated by elementary correspondences from X to Y . Elements of $\mathrm{Cor}(X, Y)$ are called *finite correspondences*.

Definition 1.57. We define $\mathrm{Cor} := \mathrm{Cor}(k)$ as the category whose objects are smooth (separated) schemes of finite type over k and whose morphisms are finite correspondences:

$$\mathrm{Hom}_{\mathrm{Cor}}(X, Y) := \mathrm{Cor}(X, Y).$$

Definition 1.58. For $X, Y \in \mathrm{Cor}$, the *tensor product* is $X \otimes Y := X \times_k Y$. This makes Cor a symmetric monoidal category.

1.5.2 The category of effective geometric motives

Definition 1.59. The category $\widehat{DM}_{\mathrm{gm}}^{\mathrm{eff}}$ is defined as the localisation of $K^b(\mathrm{Cor})$, as a tensor triangulated category, with respect to:

- *Homotopy.* For $X \in \mathrm{Sm}/k$, one inverts $p_* : [X \times \mathbb{A}^1] \rightarrow [X]$.
- *Mayer–Vietoris.* For $X = U \cup V$ (Zariski open cover), one inverts the canonical map from the cone of $[U \cap V] \rightarrow [U] \oplus [V]$ to $[X]$.

The category $DM_{\mathrm{gm}}^{\mathrm{eff}}$ of *effective geometric motives* is the pseudo-abelian completion of $\widehat{DM}_{\mathrm{gm}}^{\mathrm{eff}}$ (formally adding kernels and images of projections).

Definition 1.60. The *category of geometric motives* DM_{gm} is defined by inverting the functor $\otimes \mathbb{Z}(1)$ on $DM_{\mathrm{gm}}^{\mathrm{eff}}$; that is, there are objects $X(n)$ for $X \in DM_{\mathrm{gm}}^{\mathrm{eff}}$, $n \in \mathbb{Z}$, with

$$\mathrm{Hom}_{DM_{\mathrm{gm}}}(X(n), Y(m)) := \varinjlim_N \mathrm{Hom}_{DM_{\mathrm{gm}}^{\mathrm{eff}}}(X \otimes \mathbb{Z}(N+n), Y \otimes \mathbb{Z}(N+m)).$$

Here $\mathbb{Z}(1) := [\widetilde{\mathbb{P}^1}][-2]$ is the Tate motive and $\mathbb{Z}(n) := \mathbb{Z}(1)^{\otimes n}$ for $n \geq 0$.

1.5.3 Sites and sheaves

Definition 1.61. A *presheaf* P on a small category $\mathcal{C}$ with values in a category $\mathcal{A}$ is a functor $P : \mathcal{C}^{op} \rightarrow \mathcal{A}$.

Definition 1.62. Let X be a k -variety. A *Nisnevich cover* $\mathcal{U} \rightarrow X$ is an étale morphism of finite type such that for each separable field extension F/k of finite degree, the map $\mathcal{U}(F) \rightarrow X(F)$ is surjective.

1.5.4 Motivic complexes and the Suslin complex

Definition 1.63. 1. The category $PST := PST(k)$ of *presheaves with transfers* is the category of additive functors $\text{Cor}^{op} \rightarrow \mathbf{Ab}$.

2. The category $\text{Shv}_{\text{Nis}}(\text{Cor})$ of *Nisnevich sheaves with transfers* is the full subcategory of PST consisting of presheaves F such that, for each $X \in \text{Sm}/k$, the restriction $F|_{X_{\text{Nis}}}$ is a sheaf.

Definition 1.64. For a presheaf F on Sm/k , let $\Delta^n := \text{Spec } k[t_0, \dots, t_n]/(\sum t_i - 1)$. We define $C_n(F)(X) := F(X \times \Delta^n)$. The *Suslin complex* is

$$C_*(F)(X) : \cdots \rightarrow C_1(F)(X) \xrightarrow{d_1} C_0(F)(X), \quad d_n = \sum_i (-1)^i \delta_i^*,$$

where $\delta_i^n : \Delta^{n-1} \rightarrow \Delta^n$ is the i -th face map.

Theorem 1.65 (Voevodsky). *Let $F \in PST$ be homotopy invariant over Sm/k . Then:*

1. *The presheaf $X \mapsto H^q(X_{\text{Nis}}, F_{\text{Nis}})$ is a presheaf with transfers over k .*
2. *F_{Nis} is strictly homotopy invariant.*
3. *$F_{\text{Zar}} = F_{\text{Nis}}$ and $H^q(X_{\text{Zar}}, F_{\text{Zar}}) = H^q(X_{\text{Nis}}, F_{\text{Nis}})$.*

Proof. See [33], Ch. 3, Thm. 4.27 and Thm. 5.7, or [20], Prop. 13.9. □

Theorem 1.66 (Voevodsky's embedding theorem). *There exists a commutative diagram of exact tensor functors*

$$\begin{array}{ccc} K^b(\text{Cor}) & \xrightarrow{L} & D(\text{Shv}_{\text{Nis}}(\text{Cor})) \\ \downarrow & & \downarrow RC_* \\ DM_{\text{gm}}^{\text{eff}} & \xrightarrow{i} & DM_-^{\text{eff}} \end{array}$$

such that i is a full embedding with dense image and $RC_(L(X)) \cong C_*(X)$.*

Proof. [33], Thm. 3.2.6. □

1.5.5 Motivic cohomology

Definition 1.67. For $X \in \text{Sm}/k$ and $q \geq 0$, set $M(X) = C_*(X)$ and define

$$H^p(X, \mathbb{Z}(q)) := \text{Hom}_{DM_-^{\text{eff}}}(M_{\text{gm}}(X), \mathbb{Z}(q)[p]).$$

Proposition 1.68. For $X \in \text{Sm}/k$ and $Y \in \text{Sch}/k$ equidimensional, $\text{CH}^i(Y, j) \cong H^{2i-j}(Y, \mathbb{Z}(i))$ for $i \geq 0, j \in \mathbb{Z}$.

Proposition 1.69. For $X \in \text{Sm}/k$:

$$H^n(X, \mathbb{Z}(1)) = \begin{cases} H_{\text{Zar}}^0(X, \mathcal{O}_X^*) & n = 1, \\ \text{Pic}(X) := H_{\text{Zar}}^1(X, \mathcal{O}_X^*) & n = 2, \\ 0 & \text{otherwise.} \end{cases}$$

1.5.6 The Voevodsky category of mixed motives

The triangulated category DM is defined from DM_-^{eff} by localising the Tate motive $\mathbb{Z}(1) \in DM_-^{\text{eff}}$. Explicitly:

Definition 1.70. There exists a triangulated tensor category DM in which the functor $\mathbb{Z}(1) \otimes - : DM \rightarrow DM$ is an equivalence, together with a full faithful exact tensor functor $DM_-^{\text{eff}} \rightarrow DM$.

Note 1.71. By construction ([33], §3.2 and [2], Thm. 4.5.67), DM is a compactly generated triangulated category with compact generators $\{M(X)(n)\}_{n \in \mathbb{Z}}$, where X ranges over smooth schemes of finite type over k . This follows from the fact that the embedding $DM_{\text{gm}}^{\text{eff}} \hookrightarrow DM$ is fully faithful with dense image (Theorem 1.66), and the objects $M(X)(n)$ generate DM under triangles, shifts, and arbitrary coproducts.

1.5.7 The motivic Postnikov tower

Definition 1.72. Let $DM^{\text{eff}}(n)$ be the localisation subcategory of DM_-^{eff} generated by objects $M(X)(m)[2m]$ with $m \geq n$ and $X \in \text{Sm}/k$. For $n \in \mathbb{Z}$, we have a tower of triangulated subcategories

$$\cdots \subset DM^{\text{eff}}(n+1) \subset DM^{\text{eff}}(n) \subset \cdots \subset DM^{\text{eff}}(0) = DM_-^{\text{eff}}.$$

By Neeman's theorem (Theorem 1.31), the inclusion $i_n : DM^{\text{eff}}(n) \hookrightarrow DM$ admits a right adjoint $r_n : DM \rightarrow DM^{\text{eff}}(n)$.

Definition 1.73. For $E \in DM_-^{\text{eff}}$, the *slice tower* is

$$\cdots \rightarrow f_{n+1}E \rightarrow f_nE \rightarrow f_{n-1}E \rightarrow \cdots, \quad f_n := i_n \circ r_n.$$

The n -th slice of E is $s_n E$, defined by the distinguished triangle

$$f_{n+1}E \longrightarrow f_n E \longrightarrow s_n E.$$

Proposition 1.74. *For every $m, n \in \mathbb{Z}$,*

$$f_m(\mathbb{Z}(n)) = \begin{cases} \mathbb{Z}(n) & \text{if } m \leq n, \\ 0 & \text{if } m > n. \end{cases}$$

Equivalently, $s_n(\mathbb{Z}(n)) = \mathbb{Z}(n)$ and $s_m(\mathbb{Z}(n)) = 0$ for $m \neq n$.

Proof. Recall from Definition 1.73 that $f_m = i_m \circ r_m$, where $i_m : DM^{\text{eff}}(m) \hookrightarrow DM$ is the inclusion and $r_m : DM \rightarrow DM^{\text{eff}}(m)$ is its right adjoint. The category $DM^{\text{eff}}(m)$ is generated by objects $M(X)(j)[2j]$ with $j \geq m$ and $X \in \text{Sm}/k$.

Case $m \leq n$: We claim $\mathbb{Z}(n) \in DM^{\text{eff}}(m)$, which gives $r_m(\mathbb{Z}(n)) = \mathbb{Z}(n)$ and hence $f_m(\mathbb{Z}(n)) = i_m(\mathbb{Z}(n)) = \mathbb{Z}(n)$. Indeed, $\mathbb{Z}(n) = \mathbb{Z}(1)^{\otimes n}$ is the n -th Tate motive; by construction ([33], §3.1) it belongs to $DM^{\text{eff}}(n) \subset DM^{\text{eff}}(m)$ since $n \geq m$.

More precisely, using Proposition 2.2(4) adapted to $DM^{\text{eff}}(m)$: the counit $\epsilon_m^{\mathbb{Z}(n)} : f_m(\mathbb{Z}(n)) \rightarrow \mathbb{Z}(n)$ is an isomorphism if and only if $\mathbb{Z}(n) \in DM^{\text{eff}}(m)$. Since $\mathbb{Z}(n)$ is a generator of $DM^{\text{eff}}(n)$ and $DM^{\text{eff}}(n) \subset DM^{\text{eff}}(m)$ for $n \geq m$, we conclude $f_m(\mathbb{Z}(n)) \cong \mathbb{Z}(n)$.

Case $m > n$: We claim $\text{Hom}_{DM}(M(X)(a)[b], \mathbb{Z}(n)) = 0$ for all $X \in \text{Sm}/k$ and $a \geq m > n$, $b \in \mathbb{Z}$. By the motivic cohomology calculation (Proposition 1.69 and [33], Cor. 3.4) together with the vanishing $H^p(X, \mathbb{Z}(q)) = 0$ for $q < 0$ and for $p > 2q$ ([33], Thm. 4.3.7 and [34]), one has

$$\text{Hom}_{DM}(M(X)(a)[b], \mathbb{Z}(n)) = \text{Hom}_{DM}(M(X), \mathbb{Z}(n-a)[b]) = H^b(X, \mathbb{Z}(n-a)) = 0$$

since $n-a < 0$ (as $a \geq m > n$). Therefore $\mathbb{Z}(n)$ is orthogonal to all compact generators of $DM^{\text{eff}}(m)$, i.e. $\mathbb{Z}(n) \in DM^{\text{eff}}(m)^\perp$.

By the definition of $f_m = i_m \circ r_m$ and the adjunction $\text{Hom}_{DM}(i_m A, E) = \text{Hom}_{DM^{\text{eff}}(m)}(A, r_m E)$, taking $E = \mathbb{Z}(n)$ and $A \in DM^{\text{eff}}(m)$ arbitrary:

$$\text{Hom}_{DM}(A, f_m(\mathbb{Z}(n))) = \text{Hom}_{DM}(i_m A, \mathbb{Z}(n)) = 0,$$

since $i_m A \in DM^{\text{eff}}(m)$ is built from generators $M(X)(a)[2a]$ with $a \geq m > n$. As $DM^{\text{eff}}(m)$ is compactly generated, we conclude $f_m(\mathbb{Z}(n)) = 0$.

Slices: The distinguished triangle $f_{n+1}(\mathbb{Z}(n)) \rightarrow f_n(\mathbb{Z}(n)) \rightarrow s_n(\mathbb{Z}(n)) \rightarrow f_{n+1}(\mathbb{Z}(n))[1]$ becomes $0 \rightarrow \mathbb{Z}(n) \rightarrow s_n(\mathbb{Z}(n)) \rightarrow 0$ by the two cases, giving $s_n(\mathbb{Z}(n)) \cong \mathbb{Z}(n)$. For $m \neq n$: if $m > n$ then $f_m(\mathbb{Z}(n)) = 0 = f_{m+1}(\mathbb{Z}(n))$, so $s_m(\mathbb{Z}(n)) = 0$; if $m < n$ then $f_m(\mathbb{Z}(n)) \cong \mathbb{Z}(n) \cong f_{m+1}(\mathbb{Z}(n))$ and the triangle gives $s_m(\mathbb{Z}(n)) = 0$. $\square$

2 Results

2.1 The orthogonal tower

Throughout this section we work with motives with coefficients in $R = \mathbb{Z}[\frac{1}{p}]$, where p is the exponential characteristic of the base field k .

We recall the construction of the orthogonal tower from [28]. By Note 1.71, DM is compactly generated. Let $\mathcal{G}^{\text{eff}}$ denote the set of compact objects of the form $\{M(X)(p)\}_{p \geq 0}$ for $X \in \text{Sm}/k$. For $n \in \mathbb{Z}$, let $\mathcal{G}^{\text{eff}}(n)$ denote the set of compact objects of the form $\{M(X)(p)\}_{p \geq n}$ for $X \in \text{Sm}/k$.

Write $DM^\perp(n)$ for the triangulated orthogonal of $DM^{\text{eff}}(n)$ ([28], Def. 2.1.1.):

$$DM^\perp(n) = \left\{ E \in DM_-^{\text{eff}} : \text{Hom}_{DM^{\text{eff}}}(A, E) = 0 \text{ for all } A \in DM^{\text{eff}}(n) \right\}.$$

By ([28], Lem. 2.1.7(2)), $DM^\perp(n)$ is compactly generated. By Theorem 1.31, the inclusion $j_n : DM^\perp(n) \rightarrow DM$ admits a right adjoint

$$p_n : DM \longrightarrow DM^\perp(n),$$

which is triangulated.

Definition 2.1. Set $bc_{\leq n} := j_{n+1} \circ p_{n+1}$ and for $E \in DM$, the *orthogonal cover* of E is the canonical map $bc_{\leq n}E \rightarrow E$.

Proposition 2.2 ([28], Rem. 3.2.5, Prop. 3.2.4, Cor. 3.2.7). *Let $E \in DM$. The orthogonal cover $bc_{\leq n}E$ satisfies:*

1. $bc_{\leq n}E \in DM^\perp(n+1)$.
2. The counit $\theta_n : j_{n+1}p_{n+1} \rightarrow \text{id}$ is characterised up to unique isomorphism by: for all $F \in DM^\perp(n+1)$,

$$\theta_{n*}^E : \text{Hom}_{DM}(F, bc_{\leq n}E) \xrightarrow{\sim} \text{Hom}_{DM}(F, E).$$

3. There is a canonical natural transformation $\iota_n : bc_{\leq n} \rightarrow bc_{\leq n+1}$, and $bc_{\leq n} \circ bc_{\leq n+1} \cong bc_{\leq n}$.
4. For $E \in DM$: $\theta_n^E : bc_{\leq n}E \rightarrow E$ is an isomorphism in DM if and only if $E \in DM^\perp(n+1)$.

Proof. Part (1). By Definition 2.1, $bc_{\leq n}E = j_{n+1}(p_{n+1}E)$. Since $p_{n+1} : DM \rightarrow DM^\perp(n+1)$ maps into $DM^\perp(n+1)$ by definition, and j_{n+1} is the inclusion $DM^\perp(n+1) \hookrightarrow DM$, we have $bc_{\leq n}E \in DM^\perp(n+1)$.

Part (2). The unit-counit adjunction (j_{n+1}, p_{n+1}) gives, for any $F \in DM$ and $E \in DM$:

$$\text{Hom}_{DM}(F, j_{n+1}p_{n+1}E) \cong \text{Hom}_{DM^\perp(n+1)}(p_{n+1}F, p_{n+1}E).$$

If $F \in DM^\perp(n+1)$ then the unit $\eta_F : F \rightarrow j_{n+1}p_{n+1}F$ is an isomorphism (since p_{n+1} is right adjoint to the inclusion, and F is already in $DM^\perp(n+1)$), so

$$\mathrm{Hom}_{DM}(F, bc_{\leq n}E) = \mathrm{Hom}_{DM}(F, j_{n+1}p_{n+1}E) \cong \mathrm{Hom}_{DM}(F, E)$$

via post-composition with $\theta_n^E : bc_{\leq n}E \rightarrow E$ (the counit of the adjunction at E). This is natural in F and E , proving (2).

Part (3). The containment $DM^\perp(n+2) \subset DM^\perp(n+1)$ (since $DM^{\mathrm{eff}}(n+1) \subset DM^{\mathrm{eff}}(n+2)$ implies more objects are in the orthogonal at level $n+1$) gives a natural map between the corresponding projectors. Concretely, there is a canonical map

$$\begin{array}{ccc} bc_{\leq n}E & \xrightarrow{\iota_n} & bc_{\leq n+1}E \\ & \searrow \theta_n^E & \downarrow \theta_{n+1}^E \\ & & E \end{array}$$

defined by the universal property of part (2): since $bc_{\leq n}E \in DM^\perp(n+1)$ by part (1), there is a unique lift of $\theta_n^E : bc_{\leq n}E \rightarrow E$ through $\theta_{n+1}^E : bc_{\leq n+1}E \rightarrow E$, giving ι_n . Applying $bc_{\leq n}$ to the identity $bc_{\leq n+1}E \xrightarrow{\theta_{n+1}^E} E$ and using the universal property again gives $bc_{\leq n}(bc_{\leq n+1}E) \cong bc_{\leq n}E$.

Part (4). ($\Rightarrow$): If θ_n^E is an isomorphism then $E \cong bc_{\leq n}E \in DM^\perp(n+1)$ by part (1). ($\Leftarrow$): If $E \in DM^\perp(n+1)$, then the unit $\eta_E : E \rightarrow j_{n+1}p_{n+1}E = bc_{\leq n}E$ is an isomorphism (the adjunction unit is an isomorphism precisely for objects already in the full subcategory), and θ_n^E is its inverse. $\square$

Proposition 2.3 ([28], Thm. 3.2.12). *For each $E \in DM$, there is a commutative diagram in DM :*

$$\begin{array}{ccccccc} \cdots & \longrightarrow & bc_{\leq n}E & \xrightarrow{\iota_n} & bc_{\leq n+1}E & \longrightarrow \cdots & \longrightarrow \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}E \\ & & \searrow \theta_n^E & & \downarrow \theta_{n+1}^E & & \nearrow c \\ & & & & E & & \end{array}$$

where $\iota_n : bc_{\leq n}E \rightarrow bc_{\leq n+1}E$ is the canonical natural transformation of Proposition 2.2(3), the map $c : \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}E \rightarrow E$ is induced by the universal property of the homotopy colimit applied to the compatible system $\{\theta_n^E\}$, and all inner triangles commute.

Proof. The existence and properties of the maps ι_n follow from Proposition 2.2(3): there is a canonical natural transformation $bc_{\leq n} \rightarrow bc_{\leq n+1}$ for every $n \in \mathbb{Z}$, and these are compatible in the sense that the diagram

$$\begin{array}{ccc} bc_{\leq n}E & \xrightarrow{\iota_n} & bc_{\leq n+1}E \\ & \searrow \theta_n^E & \downarrow \theta_{n+1}^E \\ & & E \end{array}$$

commutes for each n . Indeed, both θ_n^E and $\theta_{n+1}^E \circ \iota_n$ are morphisms $bc_{\leq n}E \rightarrow E$ in DM . Since $bc_{\leq n}E \in DM^\perp(n+1)$ by Proposition 2.2(1), the universal property of θ_{n+1}^E (Proposition 2.2(2)) gives a unique factorisation of any morphism from $bc_{\leq n}E$ to E through $bc_{\leq n+1}E$ — but θ_n^E already maps to E directly, and ι_n followed by θ_{n+1}^E gives the same map, so both composites agree.

The maps $\theta_n^E : bc_{\leq n}E \rightarrow E$ thus form a compatible system indexed over $\mathbb{Z}$. By the universal property of the homotopy colimit (see Definition 1.28 and [24], Prop. 1.6.8), there is a unique morphism

$$c : \operatorname{hocolim}_{n \rightarrow \infty} bc_{\leq n}E \longrightarrow E$$

such that for every n the composite $bc_{\leq n}E \rightarrow \operatorname{hocolim}_{n \rightarrow \infty} bc_{\leq n}E \xrightarrow{c} E$ equals θ_n^E . This is precisely the statement that all inner triangles in the displayed diagram commute. $\square$

Definition 2.4. The tower of Proposition 2.3 is called the *orthogonal tower*.

Definition 2.5. For $E, F \in DM$, consider the increasing filtration $F_\bullet$ on $\operatorname{Hom}_{DM}(F, E)$ (resp. $\operatorname{Hom}_{DM}(F, \operatorname{hocolim}_{n \rightarrow \infty} bc_{\leq n}E)$) where F_p is the image of $\theta_{p*}^E : \operatorname{Hom}_{DM}(F, bc_{\leq p}E) \rightarrow \operatorname{Hom}_{DM}(F, E)$ (resp. $\lambda_{p*}^E : \operatorname{Hom}_{DM}(F, bc_{\leq p}E) \rightarrow \operatorname{Hom}_{DM}(F, \operatorname{hocolim} bc_{\leq n}E)$).

Remark 2.6 (Comparison with the slice tower). It is worth contrasting the orthogonal tower $\{bc_{\leq n}\}$ of Definition 2.1 with the slice tower $\{f_n\}$ of Definition 1.73, since both arise from the same mechanism — Neeman’s adjoint functor theorem (Theorem 1.31) applied to a nested family of localising subcategories — but run in opposite directions. The slice functor $f_n = i_n \circ r_n$ is built from the inclusion $i_n : DM^{\text{eff}}(n) \hookrightarrow DM$ of the subcategory *generated* by Tate twists of weight $\geq n$, followed by its right adjoint r_n ; thus $f_n E$ retains only the part of E visible to *high-weight* test objects, and $f_n(\mathbb{Z}(n)) \cong \mathbb{Z}(n)$ while $f_m(\mathbb{Z}(n)) = 0$ once m exceeds n (Proposition 1.74). The orthogonal cover $bc_{\leq n} = j_{n+1} \circ p_{n+1}$ is built instead from the inclusion $j_{n+1} : DM^\perp(n+1) \hookrightarrow DM$ of the subcategory *orthogonal to* weight $\geq n+1$ test objects, so $bc_{\leq n}E$ retains the part of E that cannot be detected by high-weight probes. Both towers stabilise on a Tate motive $\mathbb{Z}(m)[r]$ — the slice tower because $\mathbb{Z}(m)[r]$ eventually leaves $DM^{\text{eff}}(n)$ as n grows past m , the orthogonal tower because $\mathbb{Z}(m)[r]$ enters $DM^\perp(n+1)$ once $n \geq m$ (Example 2.10) — but for a general object, and especially for objects of $\mathcal{SH}$ such as the sphere spectrum $\mathbf{1}$, the two towers behave very differently: Proposition 2.16 below shows the orthogonal tower on $\mathbf{1} \in \mathcal{SH}$ never stabilises, precisely because $\mathbf{1}$ has nonzero slices $s_m \mathbf{1}$ in every weight $m \geq 0$, so it is never orthogonal to $\mathcal{SH}^{\text{eff}}(n+1)$ for any n .

2.2 Convergence of the orthogonal tower

2.2.1 The orthogonal spectral sequence

Let $E, F \in DM$. By ([28], Thm. 3.2.8), one constructs canonical triangulated functors

$$bc_{p/p-1} : DM \longrightarrow DM, \quad p \in \mathbb{Z},$$

fitting into distinguished triangles in DM :

$$bc_{\leq p-1} E \longrightarrow bc_{\leq p} E \longrightarrow bc_{p/p-1} E.$$

Proposition 2.7. *The orthogonal tower induces a spectral sequence of homological type*

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(F, (bc_{p/p-1} E)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(F, E),$$

with differentials $d_r : E_{p,q}^r \rightarrow E_{p-r,q-r+1}^r$, where the abutment carries the filtration $F_\bullet$ of Definition 2.5.

Similarly, applying $\mathrm{Hom}_{DM}(F, -)$ to the tower $\{bc_{\leq n} E\}$ gives a spectral sequence converging to $\mathrm{Hom}_{DM}(F, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} E)$:

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(F, (bc_{p/p-1} E)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(F, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} E). \quad (5)$$

The canonical map $c : \mathrm{hocolim} bc_{\leq n} E \rightarrow E$ of Proposition 2.3 induces a morphism of spectral sequences from (5) to the first that is the identity on E_1 -terms.

Proof. Apply the cohomological functor $\mathrm{Hom}_{DM}(F, -)$ to each distinguished triangle in the orthogonal tower:

$$bc_{\leq p-1} E \longrightarrow bc_{\leq p} E \longrightarrow bc_{p/p-1} E \longrightarrow (bc_{\leq p-1} E)[1].$$

Setting $D^{p,q} := \mathrm{Hom}_{DM}(F, (bc_{\leq -p} E)[q+p])$ and $E^{p,q} := \mathrm{Hom}_{DM}(F, (bc_{-p/-p-1} E)[q+p])$, the long exact sequences in $\mathrm{Hom}_{DM}(F, -)$ assemble into an exact couple

$$\begin{array}{ccc} \bigoplus_{p,q} D^{p,q} & \xrightarrow{\alpha} & \bigoplus_{p,q} D^{p,q} \\ & \nwarrow \gamma \quad \nearrow \beta & \\ & \bigoplus_{p,q} E^{p,q} & \end{array}$$

where:

- $\alpha : D^{p,q} \rightarrow D^{p-1,q+1}$ is induced by the tower map $\iota_{-p-1} : bc_{\leq -p-1} E \rightarrow bc_{\leq -p} E$;
- $\beta : D^{p,q} \rightarrow E^{p,q}$ is induced by the quotient map $bc_{\leq -p} E \rightarrow bc_{-p/-p-1} E$;

- $\gamma : E^{p,q} \rightarrow D^{p+1,q-1}$ is the connecting map $bc_{-p/-p-1}E \rightarrow (bc_{\leq -p-1}E)[1]$ from the distinguished triangle.

Exactness at each vertex is exactness of the long exact sequence in $\mathrm{Hom}_{DM}(F, -)$ applied to these triangles. The bidegrees $\deg \alpha = (-1, 1)$, $\deg \beta = (0, 0)$, $\deg \gamma = (1, 0)$ match Note 1.44, so Proposition 1.43 yields a spectral sequence $\{E_r^{p,q}, d_r\}_{r \geq 1}$ with

$$E_1^{p,q} = E^{p,q} = \mathrm{Hom}_{DM}(F, (bc_{p/p-1}E)[q-p]).$$

The colimit group satisfies $D^{-\infty,q} = \varinjlim_p D^{p,q} = \mathrm{Hom}_{DM}(F, (\mathrm{hocolim}_n bc_{\leq n}E)[q])$, giving abutment (5). The morphism c_* is a morphism of exact couples (it commutes with α, β, γ since c commutes with all tower maps and quotients), hence a morphism of spectral sequences, which is the identity on E_1 since c is the identity on each $bc_{p/p-1}E$ piece. $\square$

2.2.2 Orthogonality and duality

Throughout this section, let $Y \in \mathrm{Sm}/k$ be a connected scheme of dimension d and let $r, s \in \mathbb{Z}$.

Proposition 2.8. $M(Y)(s)[r] \in DM^\perp(d+s+1)$.

Proof. By ([28], 2.2), it suffices to show that $\mathrm{Hom}_{DM}(M(X)(a)[b], M(Y)(s)[r]) = 0$ for all $X \in \mathrm{Sm}/k$ and $a, b \in \mathbb{Z}$ with $a \geq d+s+1$. By ([19], Cor. 4.13, Thm. 4.12, Thm. 5.1) we may assume the base field k is perfect. If k admits resolution of singularities, by ([33], Thm. 4.3.7),

$$\mathrm{Hom}_{DM}(M(X)(a)[b], M(Y)(s)[r]) \cong \mathrm{Hom}_{DM}(M(X) \otimes M^c(Y)(e)[f], \mathbb{Z})$$

where $M^c(Y) \in DM_-^{\mathrm{eff}}$ is the motive of Y with compact support ([33], §4.1, Cor. 4.1.6), $e = a - s - d$ and $f = b - r - 2d$. For a perfect field of positive characteristic, the same conclusion follows from ([30], Thm. 5.5.14 and Lem. 5.5.6). The hypothesis $a \geq d+s+1$ implies $e = a - s - d \geq 1$, so $M(X) \otimes M^c(Y)(e)[f] \in DM^{\mathrm{eff}}(1)$.

We therefore need to show $\mathrm{Hom}_{DM}(A, \mathbb{Z}) = 0$ for $A \in DM^{\mathrm{eff}}(1)$. Since $DM^{\mathrm{eff}}(1)$ is generated by $\{M(X)(j)[2j] : j \geq 1, X \in \mathrm{Sm}/k\}$, it suffices to verify this on generators. Using the motivic cohomology isomorphism $\mathrm{Hom}_{DM}(M(X)(j)[b], \mathbb{Z}) = H^{b-2j}(X, \mathbb{Z}(-j))$ and the vanishing $H^p(X, \mathbb{Z}(q)) = 0$ for $q < 0$ ([33], Cor. 3.4), we get zero since $-j \leq -1 < 0$. Therefore $M(X) \otimes M^c(Y)(e)[f] \in DM^{\mathrm{eff}}(1)$ maps to zero in $\mathrm{Hom}_{DM}(-, \mathbb{Z})$, completing the proof. $\square$

Corollary 2.9. *Let $E = M(Y)(s)[r] \in DM$ where $Y \in \mathrm{Sm}/k$ is connected of dimension d . Then:*

1. *The natural map $\theta_{d+s}^E : bc_{\leq d+s}E \rightarrow E$ is an isomorphism in DM .*

2. For any $A \in DM$ and any $f : E \rightarrow A$ in DM , there exists a unique lifting $g : E \rightarrow bc_{\leq d+s}A$ such that the diagram

$$\begin{array}{ccc} & & bc_{\leq d+s}A \\ & \nearrow g & \downarrow \theta_{d+s}^A \\ E & \xrightarrow{f} & A \end{array}$$

commutes in DM .

3. $f = 0$ if and only if $g = 0$.

Proof. Part (1). By Proposition 2.8, $E = M(Y)(s)[r] \in DM^\perp(d+s+1)$. By Proposition 2.2(4), $\theta_{d+s}^E : bc_{\leq d+s}E \rightarrow E$ is an isomorphism if and only if $E \in DM^\perp(d+s+1)$, which holds.

Part (2). Applying part (1) with A in place of E : since $E = M(Y)(s)[r] \in DM^\perp(d+s+1)$ by Proposition 2.8, and $\theta_{d+s}^A : bc_{\leq d+s}A \rightarrow A$ satisfies the universal property of Proposition 2.2(2), we obtain for any $f : E \rightarrow A$ a unique $g : E \rightarrow bc_{\leq d+s}A$ with $\theta_{d+s}^A \circ g = f$. The diagram commutes by construction. Uniqueness follows because two lifts g, g' satisfy $\theta_{d+s}^A \circ (g - g') = 0$, and since θ_{d+s}^A is an isomorphism on $\text{Hom}_{DM}(E, -)$ by Proposition 2.2(2) (as $E \in DM^\perp(d+s+1)$), we get $g = g'$.

Part (3). If $f = 0$ then by uniqueness in part (2), $g = 0$. Conversely, if $g = 0$ then $f = \theta_{d+s}^A \circ g = 0$. $\square$

Example 2.10. Let $Y = \text{Spec } k$, so that $d = \dim Y = 0$, and let $m, r \in \mathbb{Z}$. Applying Corollary 2.9(1) to $E = \mathbb{Z}(m)[r] = M(\text{Spec } k)(m)[r]$ shows that the counit

$$\theta_m^{\mathbb{Z}(m)[r]} : bc_{\leq m}(\mathbb{Z}(m)[r]) \longrightarrow \mathbb{Z}(m)[r]$$

is already an isomorphism in DM . In other words, the orthogonal cover of a (shifted) Tate motive recovers it on the nose at stage $n = m$ of the tower — no information is lost by truncating there. This is the simplest instance of the mechanism exploited in the proof of Theorem 2.11: for $B = M(X)(s)[r]$ of dimension $d = \dim X$, the orthogonal cover $bc_{\leq d+s}A$ already computes $\text{Hom}_{DM}(B, A)$ through θ_{d+s}^A , and it is exactly this finite stage of stabilisation, uniform in A , that forces the vanishing $E_1^{p,q} = 0$ for $p \leq s-1$ used there.

2.2.3 Convergence in DM

In this section we consider objects $A, B \in DM$ where $B = M(X)(s)[r]$ for $X \in \text{Sm}/k$ and $r, s \in \mathbb{Z}$.

Theorem 2.11. *The spectral sequence*

$$E_{p,q}^1 = \text{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) \Rightarrow \text{Hom}_{DM}(B, \text{hocolim}_{n \rightarrow \infty} bc_{\leq n}A)$$

converges strongly.

Proof. We verify the three conditions of strong convergence (Definition 1.39(iii)) for the filtration $F_\bullet$ of Definition 2.5 on $H := \mathrm{Hom}_{DM}(B, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} A)$.

Exhaustiveness. Since $B = M(X)(s)[r]$ is compact in DM (Note 1.71), Proposition 1.29 gives

$$H = \mathrm{Hom}_{DM}(B, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} A) \cong \mathrm{colim}_{n \rightarrow \infty} \mathrm{Hom}_{DM}(B, bc_{\leq n} A).$$

Every element of H therefore lies in the image of some $\mathrm{Hom}_{DM}(B, bc_{\leq p} A) \rightarrow H$, i.e. in $F_p H$ for some p . Hence $F_{-\infty} H = H$.

Half-plane vanishing. By construction, $bc_{\leq n} A \in DM^\perp(n+1)$ (Proposition 2.2(1)). Since $B = M(X)(s)[r] \in DM^{\mathrm{eff}}(s)$, for $n \leq s-1$ we have $n+1 \leq s$, so $DM^{\mathrm{eff}}(s) \subset DM^{\mathrm{eff}}(n+1)$ and hence $\mathrm{Hom}_{DM}(B, bc_{\leq n} A) = 0$. Applying $\mathrm{Hom}_{DM}(B, -)$ to the triangle $bc_{\leq p-1} A \rightarrow bc_{\leq p} A \rightarrow bc_{p/p-1} A$ therefore gives $E_1^{p,q} = \mathrm{Hom}_{DM}(B, (bc_{p/p-1} A)[q-p]) = 0$ for all $p \leq s-1$. Thus the spectral sequence is confined to the half-plane $\{p \geq s\}$.

Strong convergence. For any fixed total degree $n = p+q$, only finitely many $p \geq s$ contribute non-zero $E_1^{p,q}$ once we note that $E_1^{p,q} = \mathrm{Hom}_{DM}(B, (bc_{p/p-1} A)[q-p])$ and the differentials $d_r : E_r^{p,q} \rightarrow E_r^{p-r, q-r+1}$ have source degree $p-r < p$. Therefore for r large enough all differentials into and out of position (p, q) are zero, so $E_\infty^{p,q} = E_r^{p,q}$ for $r \gg 0$, giving $RE_\infty^{p,q} = 0$. By Theorem 1.51(1) (Boardman, applied via Proposition 1.54 and Note 1.55), the spectral sequence converges strongly. $\square$

Corollary 2.12. *Suppose the canonical map*

$$c : \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} A \longrightarrow A$$

induces an isomorphism of abelian groups

$$c_* : \mathrm{Hom}_{DM}(B, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} A) \xrightarrow{\sim} \mathrm{Hom}_{DM}(B, A).$$

Then the spectral sequence

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(B, (bc_{p/p-1} A)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(B, A)$$

converges strongly.

Proof. Follows directly from Theorem 2.11 and the morphism of spectral sequences (2) in Section 2.2.1. $\square$

The following is the main result of this thesis.

Theorem 2.13. *The canonical map*

$$c : \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} A \xrightarrow{\sim} A$$

is an isomorphism in DM . Consequently, the spectral sequence

$$E_{p,q}^1 = \mathrm{Hom}_{DM}(B, (bc_{p/p-1}A)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(B, A)$$

converges strongly.

Proof. By Corollary 2.12, it suffices to show that c is an isomorphism in DM . Since DM is compactly generated, it suffices to show that for all $a, b \in \mathbb{Z}$ and all $Y \in \mathrm{Sm}/k$, the map

$$c_* : \mathrm{Hom}_{DM}(M(Y)(a)[b], \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}A) \longrightarrow \mathrm{Hom}_{DM}(M(Y)(a)[b], A)$$

is an isomorphism of abelian groups.

Surjectivity. Given $f : M(Y)(a)[b] \rightarrow A$, by Corollary 2.9(2) (with $d = \dim Y$), there is a lifting $g : M(Y)(a)[b] \rightarrow bc_{\leq d+a}A$ with $\theta_{d+a}^A \circ g = f$. Surjectivity follows from the orthogonal tower (Proposition 2.3).

Injectivity. Let $f : M(Y)(a)[b] \rightarrow \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}A$ with $c_*(f) = 0$. Since $M(Y)(a)[b]$ is compact, by Proposition 1.29,

$$\mathrm{Hom}_{DM}(M(Y)(a)[b], \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}A) \cong \mathrm{colim}_n \mathrm{Hom}_{DM}(M(Y)(a)[b], bc_{\leq n}A),$$

so f factors as $M(Y)(a)[b] \xrightarrow{f'} bc_{\leq n}A \xrightarrow{\lambda_{n*}^A} \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}A$ for some $n \geq d + a$. Applying Corollary 2.9(2) again, f' factors as

$$M(Y)(a)[b] \xrightarrow{f''} bc_{\leq d+a}A \cong bc_{\leq d+a}(bc_{\leq n}A) \longrightarrow bc_{\leq n}A.$$

By Corollary 2.9(3), $f'' = 0$ since $\theta_{d+a}^A \circ f'' = c_*(f) = 0$, using Proposition 2.2(3) and the two commuting diagrams. Hence $f = 0$. $\square$

Remark 2.14 (Significance for the Bloch–Beilinson–Murre filtration). Recall from the Introduction that the motivation for the orthogonal tower is Peláez’s construction of a finite filtration on the Chow groups $\mathrm{CH}^q(Y)_R$, obtained by evaluating $bc_{\leq \bullet}$ on the motive of a point and mapping in $M(Y)(-q)[-2q]$. Theorem 2.13 shows that, for every $B = M(X)(r)[s]$ — not merely the specific twist used to define the Chow-group filtration — the orthogonal spectral sequence converges strongly. Consequently the associated filtration $F_{\bullet}$ on $\mathrm{Hom}_{DM}(B, A)$ (Definition 2.5) is automatically exhaustive, separated, and complete for every smooth X and every twist and shift, so that $\mathrm{Hom}_{DM}(B, A)$ is recovered from the tower, up to the extension problem recorded by the E_{∞} -page. This is precisely the convergence statement needed before properties expected of a Bloch–Beilinson–Murre filtration — such as finiteness of the graded pieces $\mathrm{Gr}_F^p H$ — can be meaningfully transported from statements about the individual layers $bc_{p/p-1}A$ to statements about $\mathrm{Hom}_{DM}(B, A)$ itself: without strong convergence, the filtration could fail to be exhaustive, or the comparison between $E_{\infty}^{p,q}$ and $\mathrm{Gr}_F^p \mathrm{Hom}_{DM}(B, A)$ could carry an uncontrolled

discrepancy (Lemma 1.48). Theorem 2.13 removes this obstruction unconditionally, without assuming any part of the (still conjectural) Bloch–Beilinson–Murre programme itself.

2.3 The $\mathbb{A}^1$ -stable homotopy category of Morel–Voevodsky

We refer the reader to ([18], Thm. 4.15) for the construction of the stable model structure on symmetric T -spectra. We write $\mathcal{SH}$ for its homotopy category, the $\mathbb{A}^1$ -stable homotopy category of Morel–Voevodsky.

Let $\Sigma_T^\infty X_+ \in \mathcal{SH}$, $X \in \mathbf{Sm}/k$, be the infinite suspension of the simplicial presheaf represented by X with a disjoint basepoint. By ([18], Prop. 4.19), $\mathcal{SH}$ is a tensor triangulated category with unit $\mathbf{1} = \Sigma_T^\infty \mathrm{Spec} k_+$. Writing $E(1)$ for $E \otimes \Sigma_T^\infty(\mathbb{G}_m)[-1]$ and $E(n) = (E(n-1))(1)$ inductively for $n \geq 0$, the functor $E \mapsto E(1)$ is an autoequivalence ([16], 8.10; [2], Thm. 4.3.38). Like DM , $\mathcal{SH}$ is compactly generated with compact generators

$$\mathcal{G}_{\mathcal{SH}} = \left\{ \Sigma_T^\infty X_+(p) : X \in \mathbf{Sm}/k, p \in \mathbb{Z} \right\}.$$

2.3.1 The orthogonal tower in $\mathcal{SH}$

The construction of Section 1.4 and §1.5.7 carries over to $\mathcal{SH}$ verbatim, replacing $DM^{\mathrm{eff}}(n)$ by the analogous subcategory $\mathcal{SH}^{\mathrm{eff}}(n)$.

Definition 2.15. For $n \in \mathbb{Z}$, let $\mathcal{SH}^{\mathrm{eff}}(n)$ denote the localisation subcategory of $\mathcal{SH}$ generated by $\{\Sigma_T^\infty X_+(p) : p \geq n, X \in \mathbf{Sm}/k\}$. Define

$$\mathcal{SH}^\perp(n) := \left\{ E \in \mathcal{SH} : \mathrm{Hom}_{\mathcal{SH}}(A, E) = 0 \text{ for all } A \in \mathcal{SH}^{\mathrm{eff}}(n) \right\}.$$

By ([18], Thm. 4.15 and [28], Lem. 2.1.7), $\mathcal{SH}^\perp(n)$ is compactly generated, so by Theorem 1.31 the inclusion $j_n^{\mathcal{SH}} : \mathcal{SH}^\perp(n) \hookrightarrow \mathcal{SH}$ admits a right adjoint $p_n^{\mathcal{SH}} : \mathcal{SH} \rightarrow \mathcal{SH}^\perp(n)$. Set

$$bc_{\leq n}^{\mathcal{SH}} := j_{n+1}^{\mathcal{SH}} \circ p_{n+1}^{\mathcal{SH}} : \mathcal{SH} \longrightarrow \mathcal{SH},$$

with counit $\theta_n^{E, \mathcal{SH}} : bc_{\leq n}^{\mathcal{SH}} E \rightarrow E$.

The properties of Proposition 2.2 hold in $\mathcal{SH}$ with identical proofs (replacing DM by $\mathcal{SH}$ throughout). In particular, for each $E \in \mathcal{SH}$ there is an orthogonal tower

$$\begin{array}{ccccccc} \cdots & \longrightarrow & bc_{\leq n}^{\mathcal{SH}} E & \longrightarrow & bc_{\leq n+1}^{\mathcal{SH}} E & \longrightarrow & \cdots \longrightarrow \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}^{\mathcal{SH}} E \\ & & \searrow \theta_n^{E, \mathcal{SH}} & & \downarrow \theta_{n+1}^{E, \mathcal{SH}} & & \nearrow c^{\mathcal{SH}} \\ & & & & E & & \end{array}$$

and, for $F, E \in \mathcal{SH}$, a spectral sequence (Proposition 2.7 applied in $\mathcal{SH}$; see also [27]):

$$E_{p,q}^1 = \mathrm{Hom}_{\mathcal{SH}}(F, (bc_{p/p-1}^{\mathcal{SH}} E)[q-p]) \Rightarrow \mathrm{Hom}_{\mathcal{SH}}(F, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n}^{\mathcal{SH}} E).$$

We henceforth drop the superscript $\mathcal{SH}$ and write $bc_{\leq n}$, θ_n^E , etc. when the ambient category is clear from context.

2.3.2 Failure of strong convergence for $\mathbf{1} \in \mathcal{SH}$

Proposition 2.16. *The canonical map*

$$c : \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} \mathbf{1} \longrightarrow \mathbf{1}$$

is not an isomorphism in $\mathcal{SH}$.

Proof. Suppose for contradiction that c is an isomorphism. Since $\mathbf{1} \in \mathcal{SH}$ is compact (it is among the generators $\mathcal{G}_{\mathcal{SH}}$), Proposition 1.29 gives

$$\mathrm{colim}_{n \rightarrow \infty} \mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, bc_{\leq n} \mathbf{1}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} \mathbf{1}) \xrightarrow{c_*} \mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, \mathbf{1}).$$

Since c_* would be an isomorphism, $\mathrm{id}_{\mathbf{1}}$ is in the image of some $\mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, bc_{\leq n} \mathbf{1}) \rightarrow \mathrm{Hom}_{\mathcal{SH}}(\mathbf{1}, \mathbf{1})$, meaning $\mathrm{id}_{\mathbf{1}}$ factors as:

$$\begin{array}{ccc} & & bc_{\leq n} \mathbf{1} \\ & \nearrow \sigma & \downarrow \theta_n^{\mathbf{1}} \\ \mathbf{1} & \xrightarrow{\mathrm{id}} & \mathbf{1}. \end{array}$$

From $\theta_n^{\mathbf{1}} \circ \sigma = \mathrm{id}_{\mathbf{1}}$, we get that σ is a split monomorphism, so $\mathbf{1}$ is a direct summand of $bc_{\leq n} \mathbf{1}$. Since $bc_{\leq n} \mathbf{1} \in \mathcal{SH}^{\perp}(n+1)$ by Proposition 2.2(1) (in $\mathcal{SH}$), and $\mathcal{SH}^{\perp}(n+1)$ is closed under direct summands (it is a thick subcategory by Proposition 1.18), we conclude $\mathbf{1} \in \mathcal{SH}^{\perp}(n+1)$.

However, by ([35], Thm. 2.2(3)) $s_m \mathbf{1} = \mathbf{1}_{s_m} \neq 0$ for all $m \geq 0$; more precisely, $\mathbf{1} \in \mathcal{SH}^{\mathrm{eff}}(m) \setminus \mathcal{SH}^{\mathrm{eff}}(m+1)$ for every m ([21], Conj. 9 and p. 350), so $\mathbf{1} \notin \mathcal{SH}^{\perp}(n+1)$ for any n . This is a contradiction. $\square$

Note 2.17. The argument of Proposition 2.16 applies to any compact $A \in \mathcal{SH}$ for which $s_m A \neq 0$ for arbitrarily large m .

Corollary 2.18. *For the sphere spectrum $\mathbf{1} \in \mathcal{SH}$, the spectral sequence*

$$E_{p,q}^1 = \mathrm{Hom}_{\mathcal{SH}}(\Sigma_T^{\infty} X_+(s)[t], (bc_{p/p-1} \mathbf{1})[q-p]) \Rightarrow \mathrm{Hom}_{\mathcal{SH}}(\Sigma_T^{\infty} X_+(s)[t], \mathbf{1})$$

does not converge strongly for any $\Sigma_T^{\infty} X_+(s)[t]$ with $X \in \mathrm{Sm}/k$, $s, t \in \mathbb{Z}$.

Proof. Suppose for contradiction that the spectral sequence converges strongly for some generator $B = \Sigma_T^\infty X_+(s)[t]$. The morphism of spectral sequences induced by c_* (Proposition 2.7 applied in $\mathcal{SH}$, equation (5)) is the identity on E_1 -terms, so by Theorem 1.41 the comparison map

$$c_* : \mathrm{Hom}_{\mathcal{SH}}(B, \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} \mathbf{1}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{SH}}(B, \mathbf{1})$$

is an isomorphism of filtered groups. Since $\mathcal{SH}$ is compactly generated by $\mathcal{G}_{\mathcal{SH}}$ and this holds for every $B \in \mathcal{G}_{\mathcal{SH}}$ (by the same argument for each generator), c is an isomorphism in $\mathcal{SH}$ — contradicting Proposition 2.16. $\square$

2.3.3 Strong convergence for $\mathcal{SH}^\perp(r)$ -objects

Proposition 2.19. *Let $A \in \mathcal{SH}$ be such that $A \in \mathcal{SH}^\perp(r)$ for some $r \in \mathbb{Z}$. Then:*

1. *The canonical map $c : \mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} A \xrightarrow{\sim} A$ is an isomorphism in $\mathcal{SH}$.*
2. *The spectral sequence*

$$E_{p,q}^1 = \mathrm{Hom}_{\mathcal{SH}}(\Sigma_T^\infty X_+(s)[t], (bc_{p/p-1} A)[q-p]) \Rightarrow \mathrm{Hom}_{\mathcal{SH}}(\Sigma_T^\infty X_+(s)[t], A)$$

converges strongly for all $X \in \mathrm{Sm}/k$, $s, t \in \mathbb{Z}$.

Proof. *Part (1).* Let $m \geq r$ be arbitrary. Since $DM^{\mathrm{eff}}(m) \supset DM^{\mathrm{eff}}(r)$ is implied by $m \geq r$ (more conditions generate a larger orthogonal), and $A \in \mathcal{SH}^\perp(r)$ means $\mathrm{Hom}_{\mathcal{SH}}(B, A) = 0$ for all $B \in \mathcal{SH}^{\mathrm{eff}}(r)$, we get $A \in \mathcal{SH}^\perp(m)$ for all $m \geq r$ since $\mathcal{SH}^{\mathrm{eff}}(m) \subset \mathcal{SH}^{\mathrm{eff}}(r)$.

By Proposition 2.2(4) in $\mathcal{SH}$: since $A \in \mathcal{SH}^\perp(m+1)$ (using $m \geq r$), the counit $\theta_m^A : bc_{\leq m} A \rightarrow A$ is an isomorphism. Therefore in the tower

$$\begin{array}{ccccccc} \cdots & \longrightarrow & bc_{\leq r-1} A & \longrightarrow & bc_{\leq r} A & \longrightarrow & bc_{\leq r+1} A \longrightarrow \cdots \\ & & & & \searrow \sim & & \downarrow \sim \\ & & & & & & A \end{array}$$

all maps θ_m^A for $m \geq r$ are isomorphisms, and the tower stabilises. The homotopy colimit of a stable tower is the stable value, so $\mathrm{hocolim}_{n \rightarrow \infty} bc_{\leq n} A \cong A$ and c is an isomorphism.

Part (2). By part (1) and Corollary 2.12 (applied in $\mathcal{SH}$ with $B = \Sigma_T^\infty X_+(s)[t]$), it suffices that c_* is an isomorphism on all Hom groups from compact generators. Since B is compact and c is now an isomorphism in $\mathcal{SH}$, c_* is immediately an isomorphism. The strong convergence then follows from Theorem 2.11 applied in $\mathcal{SH}$. $\square$

Corollary 2.20. *Let $B = \Sigma_T^\infty X_+(s)[t]$ with $X \in \mathrm{Sm}/k$, $s, t \in \mathbb{Z}$. Consider the Voevodsky slice spectral sequence ([35], §7) for $G \in \mathcal{SH}$:*

$$E_1^{m,n} = \mathrm{Hom}_{\mathcal{SH}}(B, s_m G[m+n]) \Rightarrow \mathrm{Hom}_{\mathcal{SH}}(B, G).$$

The orthogonal spectral sequence

$$E_{p,q}^1 = \mathrm{Hom}_{\mathcal{SH}}(B, (bc_{p/p-1}s_m G[m+n])[q-p]) \Rightarrow \mathrm{Hom}_{\mathcal{SH}}(B, s_m G[m+n])$$

converges strongly towards the E_1 -term $E_1^{m,n} = \mathrm{Hom}_{\mathcal{SH}}(B, s_m G[m+n])$. Moreover, the differential $d_1 : E_1^{m,n} \rightarrow E_1^{m+1,n}$ of the slice spectral sequence induces a map between the orthogonal spectral sequences for consecutive slices, fitting into the commutative diagram:

$$\begin{array}{ccc} E_{p,q}^1(s_m G[m+n]) & \xrightarrow{d_r} & E_{p-r,q-r+1}^1(s_m G[m+n]) \\ d_{1*} \downarrow & & \downarrow d_{1*} \\ E_{p,q}^1(s_{m+1} G[m+1+n]) & \xrightarrow{d_r} & E_{p-r,q-r+1}^1(s_{m+1} G[m+1+n]). \end{array}$$

Proof. By ([35], Thm. 2.2(3)), $s_m G \in \mathcal{SH}^\perp(m+1)$ for every m and every $G \in \mathcal{SH}$. Shifting: $s_m G[m+n] \in \mathcal{SH}^\perp(m+1)$ as well (shifts preserve orthogonality since $\mathcal{SH}^\perp(m+1)$ is closed under the translation functor by the triangulated structure). Therefore, with $A = s_m G[m+n]$ and $r = m+1$, Proposition 2.19(2) applies and gives strong convergence.

For the morphism of spectral sequences: the differential $d_1 : E_1^{m,n} \rightarrow E_1^{m+1,n}$ of the slice spectral sequence is induced by the natural map $\partial[m+n] : s_m G[m+n] \rightarrow s_{m+1} G[m+1+n]$, where ∂ is the composition ([35], Thm. 2.2(1))

$$s_m G \xrightarrow{\sigma_m} f_{m+1} G[1] \xrightarrow{\pi_{m+1}[1]} s_{m+1} G[1].$$

Since the orthogonal tower functor $bc_{\leq n}^{\mathcal{SH}}$ is triangulated and natural (Definition 2.15 and Proposition 2.2 in $\mathcal{SH}$), post-composition with $\partial[m+n]$ defines a morphism of exact couples (in the sense of Definition 1.40), hence a morphism of spectral sequences. The commutativity of the square with d_r follows from naturality of the exact couple construction (Proposition 1.43) with respect to morphisms of the tower. $\square$

2.4 Further applications: Chow correspondences, regular homomorphisms, and incidence equivalence

The orthogonal tower $\{bc_{\leq n}\}$ and the slice tower $\{f_n\}$ studied above are not only convergence-theoretic tools: since the publication of Theorem 2.13 as [5], they have been used by Peláez and collaborators to give purely motivic characterisations of several classical *adequate equivalence relations* (in the sense of Samuel [12]) on algebraic cycles. We summarise three such applications, each of which sharpens the picture of the Bloch–Beilinson–Murre filtration discussed in §2.4.3 below and uses, verbatim, the towers $bc_{\leq n}$ and f_n constructed in §2.1 and §1.5.7 respectively. We state the results with the notation of this thesis; full proofs are in the cited

sources.

2.4.1 The orthogonal cover detects vanishing of regular homomorphisms into abelian varieties

Recall (Lieberman [8]) that a group homomorphism $\psi : \mathrm{CH}_{\mathrm{alg}}^n(X) \rightarrow A(k)$, for A an abelian variety, is a *regular homomorphism* if it is induced, family-wise, by morphisms of varieties; the prototypical example is Weil’s Abel–Jacobi map. Given $X, Y \in \mathrm{SmProj}_k$ with X of dimension d , a base point $x_0 \in X(k)$, and a Chow correspondence $\Lambda \in \mathrm{CH}^{d+n}(Y \times X)$, one obtains a cycle-induced regular homomorphism $\psi : \mathrm{CH}^{d_Y-n}(Y) \rightarrow \mathrm{Alb}(X)(k)$, $\gamma \mapsto \mathrm{alb}_X(\Lambda(\gamma) - \deg(\Lambda(\gamma))[x_0])$, into the Albanese variety of X .

Theorem 2.21 (Peláez [9]). *Let $\underline{\mathrm{Alb}}(X) \in DM$ be the functor of points of $\mathrm{Alb}(X)$ and $\mathcal{A}_f : M(X) \rightarrow \underline{\mathrm{Alb}}(X)$ the map classifying the canonical morphism $X \rightarrow \mathrm{Alb}(X)$ sending x_0 to 0. Dualising $M(X)$ gives $\Lambda_X : M(Y)(-n)[-2n] \rightarrow M(X)$ from the correspondence Λ . Then a lift Λ' of $\mathcal{A}_f \circ \Lambda_X$ through the orthogonal cover $\theta_{-1}^{\underline{\mathrm{Alb}}(X)} : bc_{\leq -1}(\underline{\mathrm{Alb}}(X)) \rightarrow \underline{\mathrm{Alb}}(X)$ exists in DM if and only if $\psi \equiv 0$.*

Proof (sketch, following [9]). By the universal property of the counit $\varepsilon_0 : f_0 \rightarrow \mathrm{id}$ of the slice-tower adjunction ([28], 5.3.2 — the analogue, for the source object, of the universal property of θ_{-1} used in Proposition 2.2(2) for the target), a lift Λ' through $\theta_{-1}^{\underline{\mathrm{Alb}}(X)}$ exists if and only if the composite

$$f_0(M(Y)(-n)[-2n]) \xrightarrow{\varepsilon_0} M(Y)(-n)[-2n] \xrightarrow{\Lambda_X} M(X) \xrightarrow{\mathcal{A}_f} \underline{\mathrm{Alb}}(X)$$

is zero in DM . Since $\mathbf{1} \in DM_-^{\mathrm{eff}}$, the counit ε_{0*} is an isomorphism on $\mathrm{Hom}_{DM}(\mathbf{1}, -)$ by the same universal property, so the displayed composite vanishes if and only if it vanishes after applying $\mathrm{Hom}_{DM}(\mathbf{1}, -)$. Under the identifications $\mathrm{Hom}_{DM}(\mathbf{1}, M(Y)(-n)[-2n]) \cong \mathrm{CH}^{d_Y-n}(Y)$ and $\mathrm{Hom}_{DM}(\mathbf{1}, \underline{\mathrm{Alb}}(X)) \cong \mathrm{Alb}(X)(k)$, the map Λ_X induces $\gamma \mapsto \Lambda(\gamma)$ (Poincaré duality) and $\mathcal{A}_f$ induces $\alpha \mapsto \mathrm{alb}_X(\alpha - \deg(\alpha)[x_0])$ (naturality of the Albanese map, using $\mathrm{CH}^d(X) \cong \mathrm{CH}_{\mathrm{alg}}^d(X) \oplus \mathbb{Z}$); composing these two identifications is exactly ψ . Hence the displayed composite vanishes on $\mathrm{Hom}_{DM}(\mathbf{1}, -)$ if and only if $\psi \equiv 0$, proving the theorem. $\square$

Two immediate corollaries make this criterion effective: if Λ is algebraically equivalent to zero *on both factors* (“square equivalent to zero”), or — over $k = \mathbb{C}$ — lies in the second step of H. Saito’s filtration [11], then the lift through $bc_{\leq -1}$ automatically exists, i.e. $\psi \equiv 0$ automatically. Thus the single orthogonal cover θ_{-1} — the very map whose stabilisation (Corollary 2.9) drives the proof of Theorem 2.13 — already detects two classically defined equivalence relations at once.

2.4.2 A motivic model for the kernel of an arbitrary regular homomorphism

Peláez [7] gives a construction, using the slice tower $\{f_n\}$ of Definition 1.73 rather than the orthogonal tower, that produces a motivic model for the kernel of *any* regular homomorphism, not only one induced by a fixed correspondence.

Theorem 2.22 (Peláez [7]). *Let $X \in \text{SmProj}_k$ of dimension d and $\psi : \text{CH}_{\text{alg}}^n(X) \rightarrow A(k)$ a regular homomorphism. There exist $M_\psi^n(X) \in DM_-^{\text{eff}}$ and a map $z_\psi : M_\psi^n(X) \rightarrow M(X)$, built from the distinguished triangle $f_{d-n+1}M(X) \rightarrow f_{d-n}M(X) \rightarrow s_{d-n}M(X)$ of Definition 1.73 and a factorisation through the slice $s_{d-n}M(X)$ dictated by ψ , such that under $\text{Hom}_{DM}(\mathbb{Z}(d-n)[2d-2n], M(X)) \cong \text{CH}^n(X)$ the image of $z_{\psi*}$ is exactly $\ker(\psi) \subseteq \text{CH}_{\text{alg}}^n(X)$. Moreover, for every $r \geq 1$ the induced map on slices $s_{d-n+r}(z_\psi) : s_{d-n+r}(M_\psi^n(X)) \rightarrow s_{d-n+r}(M(X))$ is an isomorphism, i.e. $M_\psi^n(X)$ differs from $M(X)$ only in the slice at $d-n$.*

Proof (sketch, following [7]). Let $\mathcal{T} := \bigoplus M(Y)$, summed over all triples (Λ, y_0, Y) with $Y \in \text{SmProj}_k$, $\dim Y \geq 1$, $\Lambda \in \text{CH}^n(Y \times X)$, $y_0 \in Y(k)$; each summand carries a map $\Lambda_{y_0} : M(Y) \rightarrow M(X)(-d+n)[-2d+2n]$ dual to Λ , and a map $\mathcal{A}_{\psi_{\Lambda, y_0}} : M(Y) \rightarrow \underline{A}$ induced by ψ itself. By the universal property of the slice-tower counit ε_0 applied to $\varphi_{d-n}(M(X)) := f_0(M(X)(-d+n)[-2d+2n])$, the map $(\Lambda_{y_0})_\bullet : \mathcal{T} \rightarrow \varphi_{d-n}(M(X))$ factors uniquely through a map $p : \mathcal{T} \rightarrow \varphi_{d-n}(M(X))$, and $\text{Hom}_{DM}(\mathbf{1}, \varphi_{d-n}(M(X))) \cong \text{CH}^n(X)$ with the image of p_* equal to $\text{CH}_{\text{alg}}^n(X)$ (the effective cover f_0 discards exactly the part of $\text{CH}^n(X)$ not algebraically equivalent to zero). Completing p to a distinguished triangle $\mathcal{R} \rightarrow \mathcal{T} \xrightarrow{p} \varphi_{d-n}(M(X))$ and applying the homotopy t -structure, one finds $\mathcal{T}, \varphi_{d-n}(M(X)) \in (DM_-^{\text{eff}})_{\geq 0}$, so $\mathcal{R} \in (DM_-^{\text{eff}})_{\geq -1}$ with $\mathbf{h}_{-1}(\mathcal{R})$ a birational sheaf. Since $\underline{A}$ is concentrated in homotopy degree 0, the composite $(\mathcal{A}_{\psi_{\Lambda, y_0}}) \circ r : \mathcal{R} \rightarrow \underline{A}$ factors through the truncation $\mathbf{h}_{-1}(\mathcal{R})[-1]$, defining e_ψ ; the octahedral axiom applied to the resulting commutative square then produces $M_\psi^n(X)$ (a fibre of a further factorisation of p through the cofibre of e_ψ) together with $z_\psi : M_\psi^n(X) \rightarrow M(X)$, such that exactness of $\text{Hom}_{DM}(\mathbf{1}, -)$ on the resulting triangle identifies the image of $z_{\psi*}$ with $\ker(\psi) \subseteq \text{CH}_{\text{alg}}^n(X)$ — by construction $\mathcal{T}$ tests *every* correspondence-induced instance of ψ at once. The slice-agreement statement follows because the cofibre controlling the discrepancy between $M_\psi^n(X)$ and $M(X)$ is (like $\mathbf{h}_{-1}(\mathcal{R})$) a birational sheaf, hence has vanishing higher Nisnevich cohomology and so satisfies $f_1(-) \cong 0$; applying f_m to the defining triangle then shows $f_{d-n+r}(z_\psi)$ is an isomorphism for every $r \geq 1$. $\square$

Specialising ψ recovers motivic models for several classical equivalence relations: the Abel–Jacobi kernel $\text{CH}_{AJ}^n(X)$ (taking ψ to be Weil’s Abel–Jacobi map, so $A = J_a^n(X)$ over $k = \mathbb{C}$); the incidence kernel $\text{CH}_{\text{inc}}^n(X)$ (taking ψ to be the Néron–Severi regular homomorphism onto $\text{Pic}^n(X)$ of Grothendieck–Kleiman and H. Saito [11]); and, applying the same construction to the product $\mathfrak{Ab} = \prod_{\psi, A} \underline{A}$ ranging over *all* regular homomorphisms out of $\text{CH}_{\text{alg}}^n(X)$, a motivic model $M_{\text{ab}}^n(X)$

computing the subgroup $\mathrm{CH}_{\mathrm{ab}}^n(X) \subseteq \mathrm{CH}_{\mathrm{alg}}^n(X)$ of cycles *abelian equivalent to zero* (Samuel [12]) — the intersection of the kernels of every regular homomorphism from X into every abelian variety — with the same slice-agreement property above degree $d - n$.

2.4.3 Computing the second step of the orthogonal filtration: incidence equivalence

Recall from the Introduction that evaluating the orthogonal tower $bc_{\leq \bullet}$ on $\mathbb{Z}_R$ and mapping in $M(X)(-n)[-2n]$ produces the filtration $F^\bullet \mathrm{CH}^n(X)_R$ used to approach the Bloch–Beilinson–Murre conjectures — this is precisely the filtration $F_\bullet$ of Definition 2.5 applied to $B = M(X)(-n)[-2n]$, $A = \mathbb{Z}_R$. By Theorem 2.13 this filtration is unconditionally strongly convergent; Peláez [28] further shows (Theorem 2.25 above) that it satisfies BBM1, BBM2, BBM3, BBM5. Peláez [10] computes its *second step* F^2 explicitly, in terms of Griffiths’ incidence equivalence [6]: $\alpha \in \mathrm{CH}^n(X)_R$ is *incidence equivalent to zero* if $\alpha \in \mathrm{CH}_{\mathrm{alg}}^n(X)_R$ and $\pi_{Y*}((\pi_X^* \alpha) \cdot \beta) = 0 \in \mathrm{CH}^1(Y)_R$ for every $Y \in \mathrm{SmProj}_k$ and every $\beta \in \mathrm{CH}^{d-n+1}(X \times Y)_R$.

Theorem 2.23 (Peláez [10]). *Let $X \in \mathrm{SmProj}_k$ of dimension d .*

1. *For zero-cycles ($n = d$) and an arbitrary base field k : $F^2 \mathrm{CH}^d(X)_{\mathbb{Q}} = \mathrm{CH}_{\mathrm{inc}}^d(X)_{\mathbb{Q}}$; if moreover k is algebraically closed, this equals the Albanese kernel $T(X)_{\mathbb{Q}}$.*
2. *For $\alpha \in \mathrm{CH}_{\mathrm{alg}}^n(X)_{\mathbb{Q}}$, $\beta \in \mathrm{CH}_{\mathrm{alg}}^m(Y)_{\mathbb{Q}}$, both the exterior product $\alpha \otimes \beta$ and (when $Y = X$) the intersection product $\alpha \cdot \beta$ lie in F^2 of the corresponding Chow group.*
3. *If k is finite or the algebraic closure of a finite field, the same identification $F^2 \mathrm{CH}_{\mathrm{alg}}^n(X)_{\mathbb{Q}} = \mathrm{CH}_{\mathrm{inc}}^n(X)_{\mathbb{Q}}$ holds in every codimension n , not only $n = d$.*

Proof (sketch, following [10]). For part (1): if $d = 1$, $\mathrm{CH}_{\mathrm{inc}}^1(X)_{\mathbb{Q}} = 0$ (a divisor pairs nontrivially against the diagonal unless it already vanishes), and $F^2 \mathrm{CH}^1(X)_{\mathbb{Q}} = 0$ follows directly from Theorem 2.25, so both sides agree. For $d \geq 2$: the inclusion $F^2 \subseteq \mathrm{CH}_{\mathrm{inc}}^d(X)_{\mathbb{Q}}$ is the functoriality of $F^\bullet$ (Theorem 2.25) together with the defining vanishing property of the tower. For the reverse inclusion, field-extension compatibility lemmas (analogous in spirit to the reduction to a perfect, then algebraically closed, field used in Proposition 2.8) reduce to k algebraically closed. If k is not the algebraic closure of a finite field, the theory of divisorial correspondences identifies the vanishing condition defining $\mathrm{CH}_{\mathrm{inc}}^d(X)_{\mathbb{Q}}$ with membership in the Albanese kernel, and one checks $\alpha \in F^2$ directly by dimension induction (curves, threefolds, and surfaces are treated as separate base cases). If k is the algebraic closure of a finite field, algebraic and homological equivalence coincide on zero-cycles, so α additionally satisfies a second vanishing condition studied by Bloch (a K_1 -analogue of incidence equivalence, pairing

against $\mathrm{CH}^{d-n+1}(X \times Y, 1)_R$ rather than $\mathrm{CH}^{d-n+1}(X \times Y)_R$, and one concludes via the vanishing of the relevant K -cohomology of finite fields. The identification $\mathrm{CH}_{\mathrm{inc}}^d(X)_{\mathbb{Q}} = T(X)_{\mathbb{Q}}$ for k algebraically closed is the classical fact that the vanishing pairing defining incidence equivalence characterises the Albanese kernel on zero-cycles.

For part (2): reduce to k algebraically closed, write α, β as pushforwards along correspondences $\Lambda_{\alpha}, \Lambda_{\beta}$ from zero-cycles $\gamma_{\alpha}, \gamma_{\beta}$ on curves C, C' representing the Jacobians through which α, β factor; functoriality of $F^{\bullet}$ reduces the exterior product $\alpha \otimes \beta$ to $\gamma_{\alpha} \otimes \gamma_{\beta} \in \mathrm{CH}_{\mathrm{alg}}^2(C \times C')_{\mathbb{Q}}$, which lies in F^2 by part (1) once one checks it lies in the Albanese kernel of $C \times C'$ — which follows from rigidity, since the exterior product of Jacobian points factors through $J(C)(k) \times \{0\}$ and $\{0\} \times J(C')(k)$, on which the Albanese map of $C \times C'$ vanishes. The intersection product statement follows by pulling back along the diagonal $\Delta_X : X \rightarrow X \times X$ and using functoriality again.

Part (3) follows the same reduction to zero-cycles as part (1), but with the extra arithmetic input that over the algebraic closure of a finite field the vanishing of Bloch's pairing (rather than only the incidence pairing) is already implied by algebraic equivalence, so the two characterising conditions of part (1) become available in every codimension, not only for zero-cycles. $\square$

Since the filtration $F^{\bullet}$ of Theorem 2.23 is the same filtration whose strong convergence is Theorem 2.13, this result gives an unconditional, classically meaningful description of the second graded piece $\mathrm{Gr}_F^1 = F^1/F^2$ — using that F^1 itself is identified with numerical equivalence ([28], 6.1.4, recalled as BBM2 in Theorem 2.25; see also [10]) — sitting between numerical and incidence equivalence. It is complementary to, rather than a resolution of, the BBM4/BBM4a discussion of §2.4.3: it does not touch Peláez's Conjecture 2.26 on the multiplicativity of the tower, but it strengthens the case that $F^{\bullet}$ is a reasonable candidate for a Bloch–Beilinson filtration, since its first two steps are now *both* identified with classical, independently defined equivalence relations (numerical, respectively incidence, equivalence) rather than only characterised abstractly via $bc_{\leq \bullet}$.

Example 2.24. Take $k = \mathbb{C}$ and $X = E \times E'$ a product of two elliptic curves, so $d = 2$ and $\mathrm{Alb}(X) = E \times E'$. By Theorem 2.23(1), $F^2\mathrm{CH}^2(X)_{\mathbb{Q}} = T(X)_{\mathbb{Q}}$, the Albanese kernel of X . Since $H^0(X, \Omega_X^2) \cong H^0(E, \Omega_E^1) \otimes H^0(E', \Omega_{E'}^1) \neq 0$, Mumford's theorem [22] — recalled at the very start of this thesis as the original motivation for the Bloch–Beilinson–Murre programme — shows that $T(X)_{\mathbb{Q}}$ is not finitely generated: it is “enormous,” with no algebraic parametrisation. Thus the second step of the very filtration whose strong convergence is Theorem 2.13 is already, for this simple X , a group with no hope of being finitely generated. This is a concrete illustration of why the Bloch–Beilinson–Murre programme asks for a filtration that is exhaustive, separated, and complete (Theorem 2.13) — recovering $\mathrm{CH}^n(X)$ up to extensions from possibly infinite graded pieces — rather than for a

single Néron-Severi-type finiteness statement, which Mumford’s theorem already rules out beyond codimension 1.

Further results, examples, and open problems

The Bloch–Beilinson–Murre axioms: what is known, and what Theorem 2.13 does and does not add

We first make precise the remark following Theorem 2.13 (Remark 2.14) by recalling exactly which of the classical Bloch–Beilinson–Murre axioms are already established for the filtration built from the orthogonal tower, and by isolating the one that remains conditional — since this is the natural place where the convergence results of this thesis could, in principle, be brought to bear.

Following Peláez ([28], Def. 6.1.1), for $X \in \text{Sm}/k$ projective a descending filtration $F^\bullet$ on $\text{CH}^q(X)_\mathbb{Q}$ is of *Bloch–Beilinson–Murre type* if:

$$\text{BBM1 } F^0 \text{CH}^q(X)_\mathbb{Q} = \text{CH}^q(X)_\mathbb{Q};$$

$$\text{BBM2 } F^1 \text{CH}^q(X)_\mathbb{Q} = \text{CH}_{\text{num}}^q(X)_\mathbb{Q}, \text{ the cycles numerically trivial modulo rational equivalence;}$$

BBM3 $F^\bullet$ is functorial for correspondences between smooth projective varieties;

BBM4 for a correspondence $\gamma \in \text{CH}^q(X \times Y)_\mathbb{Q}$ inducing 0 on a fixed Weil cohomology H in the appropriate degree, the induced map $\text{Gr}_F^n(\text{CH}^q)$ vanishes ([28], 6.1.1);

$$\text{BBM5 } F^{q+1} \text{CH}^q(X)_\mathbb{Q} = 0.$$

Peláez also isolates a weaker substitute for BBM4 ([28], 6.1.2):

BBM4a if $\gamma \in \text{CH}^q(X \times Y)_\mathbb{Q}$ is *homologically* trivial (a strictly weaker hypothesis than in BBM4), then $\text{Gr}_F^n(\text{CH}^q) = 0$ for every $0 \leq n \leq q$.

Theorem 2.25 (Peláez, [28], Thm. 6.1.4 and Rem. 6.1.5). *The filtration on $\text{CH}^q(X)_R$ obtained by evaluating the orthogonal tower $bc_{\leq \bullet}$ on $\mathbb{Z}_R$ and mapping in $M(X)(-q)[-2q]$ (as recalled in the Introduction) satisfies BBM1, BBM2, BBM3, and BBM5, for every coefficient ring R , not only $R = \mathbb{Q}$.*

Condition BBM4 itself is, in Peláez’s words, “in general not much can be said at the moment” ([28], 6.1.6). What is known is the following conditional result towards the weaker axiom BBM4a.

Conjecture 2.26 (Peláez, [28], Conj. 6.1.7). *For all $n, m \geq 0$ there exist maps in DM*

$$\mu_{n,m} : (bc_{\leq n} \mathbb{Z}_\mathbb{Q}) \otimes (bc_{\leq m} \mathbb{Z}_\mathbb{Q}) \longrightarrow bc_{\leq n+m} \mathbb{Z}_\mathbb{Q}$$

compatible with the counits, i.e. such that $\theta_{n+m} \circ \mu_{n,m} = \theta_n \otimes \theta_m$ under the identification $\mathbb{Z}_\mathbb{Q} \otimes \mathbb{Z}_\mathbb{Q} \cong \mathbb{Z}_\mathbb{Q}$.

Proposition 2.27 (Peláez, [28], Prop. 6.1.8). *If the maps $\mu_{n,1}$ of Conjecture 2.26 exist for all $n \geq 0$, then the filtration of Theorem 2.25 satisfies BBM4a.*

Conjecture 2.26 is a statement about *multiplicative* compatibility of the tower — that the orthogonal covers $bc_{\leq \bullet}$ can be made to interact with the tensor structure of DM — and it is logically independent of the *convergence* statement proved in this thesis. Theorem 2.13 guarantees that, for every $B = M(X)(r)[s]$, the map $c : \mathrm{hocolim}_n bc_{\leq n} A \rightarrow A$ is an isomorphism and the associated filtration on $\mathrm{Hom}_{DM}(B, A)$ is exhaustive, separated and complete; it says nothing about whether the covers $bc_{\leq n}$ respect $\otimes$. Indeed one can check directly that the argument proving Theorem 2.13 never uses the tensor structure of DM — it only uses that B is compact and orthogonal to $DM^{\mathrm{eff}}(n)$ for $n \gg 0$ (Proposition 2.8) — so it cannot, by itself, produce the maps $\mu_{n,m}$ of Conjecture 2.26.

What Theorem 2.13 *does* contribute to this circle of ideas is the following: it shows that the exhaustive/separated/complete behaviour of the filtration is not special to the pairing $\mathrm{Hom}_{DM}(\mathbb{Z}_R, \mathbb{Z}_R)$ used in BBM1–BBM5, but holds uniformly for $\mathrm{Hom}_{DM}(B, A)$ with B *any* shifted Tate twist of a smooth motive. In particular, strong convergence supplies, for every such B and every A , a strongly convergent spectral sequence $E_1^{p,q} = \mathrm{Hom}_{DM}(B, (bc_{p/p-1} A)[q-p]) \Rightarrow \mathrm{Hom}_{DM}(B, A)$ (Theorem 2.13) against which any candidate multiplicative structure $\mu_{n,m}$ could be tested for compatibility with differentials, once constructed by other means. This suggests the following concrete question, left open by this thesis:

*Can the strong convergence of Theorem 2.13 be used to **construct** the maps $\mu_{n,m}$ of Conjecture 2.26 — for instance as edge homomorphisms of a multiplicative refinement of the spectral sequence of Proposition 2.7, built from a pairing $bc_{p/p-1} \otimes bc_{q/q-1} \rightarrow bc_{(p+q)/(p+q-1)}$ on associated graded pieces — or does it instead clarify an obstruction to their existence? A first, more modest test case would be to verify Conjecture 2.26 for $\mu_{n,0}$ and $\mu_{0,m}$ (which should follow formally from Proposition 2.2(3)), and then to examine whether the half-plane vanishing used in the proof of Theorem 2.11 constrains the bidegree in which $\mu_{n,1}$, if it exists, must land.*

An effective bound on the filtration length

The results already established in §2.2 in fact answer, unconditionally, the natural question of *how large* the filtration $F_\bullet$ on $\mathrm{Hom}_{DM}(B, A)$ can be.

Proposition 2.28 (Effective bound on the filtration length). *Let $B = M(X)(s)[r]$ with $X \in \mathrm{Sm}/k$ of dimension d , and let $A \in DM$ be arbitrary. Then, on $H := \mathrm{Hom}_{DM}(B, A)$, the filtration $F_\bullet$ of Definition 2.5 satisfies*

$$F_p H = 0 \quad \text{for } p \leq s - 1, \quad F_p H = H \quad \text{for } p \geq d + s.$$

In particular $\mathrm{Gr}_F^p H = 0$ unless $s \leq p \leq d + s$: the filtration has length at most $d + 1$, depending only on $\dim X$ and not on A .

Proof. The vanishing $F_p H = 0$ for $p \leq s - 1$ is already established in the proof of Theorem 2.11 (“Half-plane vanishing”): $\mathrm{Hom}_{DM}(B, bc_{\leq n} A) = 0$ for $n \leq s - 1$.

For the upper bound, apply Corollary 2.9(2)-(3) with $Y = X$ (so $d = \dim Y$) to $E = B = M(X)(s)[r]$: every $f : B \rightarrow A$ admits a unique lift $g : B \rightarrow bc_{\leq d+s} A$ with $\theta_{d+s}^A \circ g = f$, and $f = 0$ iff $g = 0$. This says precisely that $\theta_{(d+s)*}^A : \mathrm{Hom}_{DM}(B, bc_{\leq d+s} A) \rightarrow H$ is bijective, so in particular $F_{d+s} H = H$.

For $n \geq d + s$, the triangle of Proposition 2.3 gives $\theta_n^A \circ (\iota_{n-1} \circ \cdots \circ \iota_{d+s}) = \theta_{d+s}^A$, so θ_{n*}^A factors the already-surjective map $\theta_{(d+s)*}^A$; a map through which a surjection factors is itself surjective, so θ_{n*}^A is surjective and $F_n H = H$ for every $n \geq d + s$. $\square$

Example 2.29. Two extreme cases illustrate Proposition 2.28.

- $d = 0$ (so $B = \mathbb{Z}(s)[r]$, e.g. $X = \mathrm{Spec} k$): the filtration has length 1, jumping directly from $F_{s-1} H = 0$ to $F_s H = H$. This recovers, for general A , the phenomenon already observed for $A = B$ itself in Example 2.10.
- $d = 1$ (X a smooth projective curve): the filtration has length at most 2, so $H = \mathrm{Hom}_{DM}(B, A)$ is built from at most two nonzero graded pieces, $\mathrm{Gr}_F^s H$ and $\mathrm{Gr}_F^{s+1} H$, glued by a single extension problem — no matter how complicated A is.

In both cases the bound depends only on $\dim X$, uniformly in A ; this is the promised quantitative form of the vanishing used throughout §2.2. (It bounds only the *length* of the filtration on the abutment, not the page at which the spectral sequence itself degenerates termwise — that finer question is left open below.)

Coefficients: where inverting the characteristic is used

Theorem 2.13 is stated, following [28], with coefficients in $R = \mathbb{Z}[\frac{1}{p}]$, p the exponential characteristic of k . We show this hypothesis is used at exactly one point of the argument, and can be dropped whenever k is already perfect.

Proposition 2.30 (Integral statement over a perfect field). *If k is perfect, Theorem 2.13, Theorem 2.11, and Proposition 2.8 all hold with $\mathbb{Z}$ -coefficients, i.e. without inverting p .*

Proof. Inspect the proof of Proposition 2.8. Its first step invokes ([19], Cor. 4.13, Thm. 4.12, Thm. 5.1) to reduce to the case of a perfect base field; this citation is Kelly’s comparison theorem between motives over k and over a perfect closure of k , and it is precisely this comparison — needed only to move from an arbitrary k to a perfect one — that requires p to be invertible in the coefficient ring. When k is already perfect, this reduction step is vacuous: there is nothing to compare, and the proof proceeds directly from “If k admits resolution of singularities, by ([33], Thm. 4.3.7) ...” or, in positive characteristic, directly from ([30], Thm. 5.5.14 and Lem. 5.5.6) — neither of which inverts p — to the vanishing of $\mathrm{Hom}_{DM}(A, \mathbb{Z})$ for $A \in DM^{\mathrm{eff}}(1)$, which is checked integrally on generators via the vanishing of

motivic cohomology in negative weights. Hence Proposition 2.8 holds integrally once k is perfect. Since Theorem 2.11 and Theorem 2.13 are formal consequences of Proposition 2.8 together with the general machinery of §2.1–§1.4.5, none of which refers to the coefficient ring R at all, the same conclusion follows for both. $\square$

Thus the coefficient hypothesis $R = \mathbb{Z}[\frac{1}{p}]$ is needed only to cover imperfect base fields, and only through Kelly’s comparison theorem. Whether that comparison theorem itself extends to integral coefficients over an imperfect field is a question about DM over imperfect fields, independent of the convergence circle of ideas developed in this thesis, and remains open here.

Combining the orthogonal and slice spectral sequences

Corollary 2.20 converges strongly to each individual E_1 -term $\mathrm{Hom}_{\mathcal{SH}}(B, s_m G[m+n])$ of the Voevodsky slice spectral sequence. We make precise exactly how much can be proved, with the methods of this thesis, towards assembling this into a single spectral sequence for $\mathrm{Hom}_{\mathcal{SH}}(B, G)$ itself — correcting what would otherwise be an overstatement of the strength of these methods.

Proposition 2.31 (The outer, slice-indexed spectral sequence). *Let $G \in \mathcal{SH}$ and $B = \Sigma_T^\infty X_+(s)[t]$, $X \in \mathrm{Sm}/k$. Applying $\mathrm{Hom}_{\mathcal{SH}}(B, -)$ to the slice tower $\{f_m G\}_{m \in \mathbb{Z}}$ (Definition 1.73) via Proposition 1.54 produces, unconditionally, a spectral sequence*

$$E_1^{m,n} = \mathrm{Hom}_{\mathcal{SH}}(B, s_m G[m+n]) \Rightarrow \mathrm{Hom}_{\mathcal{SH}}(B, \mathrm{hocolim}_{m \rightarrow -\infty} f_m G).$$

By Corollary 2.20, each individual term $E_1^{m,n}$ is itself the strongly convergent abutment of the orthogonal spectral sequence on $s_m G[m+n]$.

Proof. Since B is a compact generator of $\mathcal{SH}$ ($\mathcal{SH}$ is compactly generated with compact generators $\mathcal{G}_{\mathcal{SH}}$, §2.3), this is a direct instance of Proposition 1.54 applied to the tower $A_n := f_{-n} G$ (with structure maps the slice-tower maps of Definition 1.73 and associated graded $Q_n = s_{-n} G$), whose hypotheses require nothing beyond compactness of B . The identification of each $E_1^{m,n}$ with a strongly convergent abutment is Corollary 2.20. $\square$

If, in addition, the slice tower converges — i.e. the canonical map $\mathrm{hocolim}_{m \rightarrow -\infty} f_m G \rightarrow G$ is an isomorphism, a hypothesis about the slice filtration alone that this thesis does not address — the abutment of Proposition 2.31 becomes $\mathrm{Hom}_{\mathcal{SH}}(B, G)$, and the outer spectral sequence converges *weakly* to it (Theorem 1.51(2)). It does *not* follow that this convergence is strong: strong convergence of the tower $\{f_{-n} G\}_{n \geq 0}$ would require the derived-limit term $RE_\infty^{m,n}$ of Lemma 1.48 to vanish, which in every strongly convergent case established in this thesis (Theorem 2.11, Proposition 2.19) is deduced from a half-plane vanishing coming from the source object B being orthogonal to $DM^{\mathrm{eff}}(n)$ (resp. $\mathcal{SH}^{\mathrm{eff}}(n)$) for $n \gg 0$ (Note 1.55).

The slice tower has no counterpart of this vanishing: the slices $s_m G$ need not vanish for $m \gg 0$ (indeed $s_m \mathbf{1} \neq 0$ for every $m \geq 0$, Proposition 2.16), so the half-plane argument that proves strong convergence throughout §2.2 and §2.3 does not apply to the outer tower here. Consequently, unlike every other result in this section, assembling the orthogonal and slice spectral sequences into a single *strongly* convergent spectral sequence for $\mathrm{Hom}_{\mathcal{SH}}(B, G)$ is **not** established by this thesis and remains open; what Proposition 2.31 does establish, unconditionally, is the weaker (but nontrivial) fact that the two constructions fit into a single formal framework, with the inner strong convergence of Corollary 2.20 feeding into each page of the outer one.

Necessity of the hypothesis in $\mathcal{SH}$

Proposition 2.19 shows that $A \in \mathcal{SH}^\perp(r)$ for some $r \in \mathbb{Z}$ is *sufficient* for strong convergence on A , and Proposition 2.16 shows the hypothesis is not vacuous, since it already fails for the unit $\mathbf{1}$. Of the questions recorded in this section, this is the one we have been least able to make progress on: unlike the filtration-length bound above, membership in some $\mathcal{SH}^\perp(r)$ is a condition on *all* weights of A at once, and the argument of Proposition 2.19(1) (that the tower stabilises once every θ_m^A , $m \geq r$, is simultaneously an isomorphism) does not obviously admit a partial or asymptotic version. We leave open whether membership in some $\mathcal{SH}^\perp(r)$ is also necessary for strong convergence, or whether a strictly larger class — for instance objects whose slices $s_m A$ vanish outside a sparse, but unbounded, set of weights — already suffices.

Concluding remarks

The convergence of the orthogonal spectral sequence, established unconditionally in Theorem 2.13 for every $B = M(X)(r)[s]$, was the central question motivating this thesis. The results assembled since — the failure and partial recovery of convergence in $\mathcal{SH}$ (§2.3), the explicit filtration-length bound and its worked examples, the precise isolation of where integral coefficients are available (Proposition 2.30), the unconditional outer spectral sequence of Proposition 2.31, and the three applications of §2.4 identifying the first two steps of the orthogonal filtration on Chow groups with numerical and incidence equivalence — together show that the convergence statement proved here is not an isolated technical fact, but the load-bearing property that makes the orthogonal tower usable as a Bloch–Beilinson–Murre filtration in the first place. What remains open — Peláez’s multiplicativity Conjecture 2.26 towards BBM4a, necessity of the $\mathcal{SH}^\perp(r)$ hypothesis, and strong convergence of the slice tower itself — is left open deliberately: each is a genuine unsolved problem in the current literature, and none is resolved by the methods developed here.

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